Addis Ababa City Administration Education Bureau

Summer Special Tutorial Program

Condensed Physics Curriculum from Grade 9-11Designed for Grade 11 Students

July 2026 G.C

Addis Ababa/Finfinne

Preface

Dear Teachers,

Addis Ababa City Administration Education Bureau has prepared this Condensed physics Curriculum for students attending the summer special tutorial program (summer camp). The curriculum focuses on core lesson notes, selected content and learning activities aligned with the syllabus. In addition, it includes key examination questions to be covered during this instructional period.

As subject experts, you are encouraged to modify the materials, provide additional explanations and include examples that you believe will help students better understand the concepts, particularly in areas where they commonly experience difficulties. You are also encouraged to draw on your own teaching experience and employ a variety of instructional strategies to keep students actively engaged throughout the program.

Overall, the program emphasizes extensive student practice, active participation, and continuous reinforcement of key concepts. It is designed to ensure that students complete each session with greater confidence and improved competence, thereby enhancing their preparedness for the coming academic year.

Lesson 16(45 minute)

Topic: Electrostatics

Sub-topics: Electric Flux, Electric Potential Energy, Electric Potential, and Equipotential Surfaces

At the end of this lesson delivery, students will be able to:

- Define electric flux conceptually and mathematically, and compute it for planar surfaces in uniform fields.
- Distinguish between the vector area orientation (normal vector) and the physical tilt of a surface relative to electric field lines.
- Define electric potential energy (U) and electric potential (V), relating them to work done by conservative electrostatic forces.
- Calculate the electric potential of point charges and relate uniform electric fields to spatial potential drops.
- Describe the geometric properties of equipotential lines/surfaces and their relationship to electric field lines.

16.1 Electric flux

- **Electric flux** measures the total number of electric field lines passing through a given surface area.
- For a uniform electric field (E) passing through a flat surface of area (A), it is a **scalar** quantity defined by the dot product:

$$\emptyset = \boldsymbol{E}.\ A = EA\cos\theta \text{ (in N.m}^2\text{/C)}$$

Where **θ** = is the angle between the **electric field vector** **E** and the **normal vector** $\hat{\mathbf{n}}$ **perpendicular** to the surface, **not** the angle between the field and the surface itself.

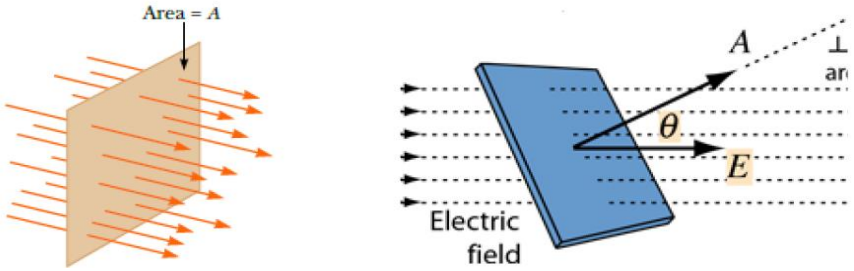


Fig 16.1 Electric flux

Activity 16.1: Hold a flat sheet of cardstock up in front of the class to represent a plane area. Use a colored pointer to represent the electric field vector. Ask students to show with their hands the direction of the normal vector n and state when the flux calculation requires.

Example 16.1: A planar square surface with an area of 0.300 m^2 is immersed in a uniform electric field of magnitude 20 N/C. The normal to the square surface forms an angle of $\varphi = 45°$ with the electric field. i) Calculate the electric flux (Φ_E) through the surface. ii) Why is the electric flux independent of the geometrical shape of the surface?

Solution: $\emptyset = \boldsymbol{E}. A = EA\cos\theta = 4.24\ N/m^2$

Exercise 16.1 Consider a plane surface of area 50 mm^2 in a uniform electric field of magnitude 25 N/C. The angle between the electric field and the normal to the surface is φ. Determine the values of angle φ for which the electric flux magnitude through the surface is i) maximum or ii) minimum.

16.2 Electric Potential Energy (U) and electric potential(V)

- **Electric potential energy:** The mechanical energy stored when work is performed against conservative electrostatic forces to bring interacting charges together. For two interacting point charges q_1 and q_2:
- Similar to a mass storing gravitational potential energy in a gravitational field of another mass.

$$U = \frac{K\,Qq}{r}$$

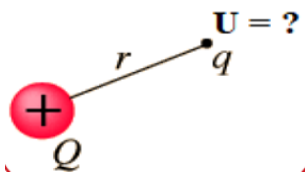


Fig 16.2 Electric potential energy

- Work done by an external force changes the electrical potential energy of a charge.

$$W = \Delta U = \frac{K\,Qq}{r}$$

- Moving a + charge toward another + charge → **stored potential energy increases.**
- Moving a + charge toward a – charge → **potential energy decreases**.

Electric potential(V): is the work done per unit charge to bring a small positive test charge from **infinity** to that point.

$$V = \frac{W_{\infty \to r}}{q} = \frac{\Delta U}{q}$$

- SI unit of electric potential is volt (V), which equals joule per coulomb (J/C).

- Charges move from high potential to low potential **(like water flowing downhill**)

Electric potential due to a point charge: is the amount of electric potential (energy per unit charge) created at a **distance r** from that single point charge.

$$V = \frac{W}{q} = \frac{KQ}{r}$$

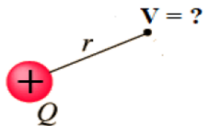


Fig 16.3 Electric potential

Potential difference in a uniform electric field: In a uniform electric field (**E**), the potential difference (**ΔV**) between two points separated by distance **d** along the direction of the field is:

$$\Delta V = Ed \rightarrow E = \Delta V/d$$

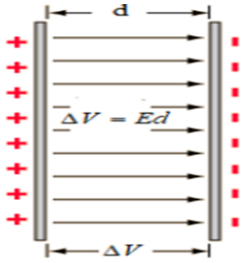


Figure 16.4: Potential difference in a uniform electric field

- The unit of electric field to be volt per meter i.e. 1 N / C =1 V / m.
- Uniform field → straight parallel lines → constant potential drop per meter.
- Potential decreases along in the field lines direction, increases against the field lines.

16.3 Equipotential line or surface

- **Equipotential Line**: A line along which the electric potential is the same at every point (2D case).
- **Equipotential Surface**: A surface along which the electric potential is the same at every point (3D case).

➢ The characteristics of equipotential lines/surfaces:

- Equipotential lines/surfaces are always perpendicular to the electric field lines.
- No work is required to move a charge along an equipotential surface: **W = q ΔV = 0**.
- The equipotential lines of a point charge are **concentric sphere** in a uniform electric field
- Two equipotential surfaces never intersect each other.

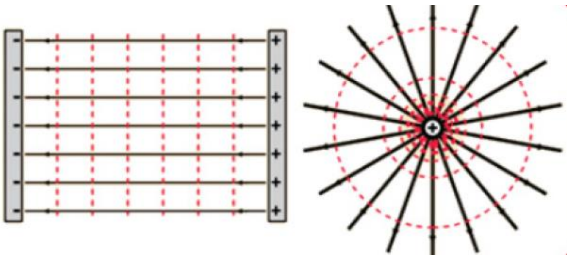


Fig 16.5 Equipotential lines of (a) a uniform field, (b) a point charge.

Example 16.2. (a) What is the electric potential energy of two electrons separated by 2.00 nm? (b) If the separation increases, does the potential energy increase or decrease?

Solution: Since the charge of an electron is '-e' = -1.6 x10^{-19}C

a. $U = \frac{K\,Qq}{r} = 1.15 \times 10^{-19}$ J b. So the potential energy decreases as r increases

Exercise16.2. Two tiny metal spheres A and B of mass m_A = 5.00 g and m_B = 10.0 g have equal positive charges q = 5.00 µC. The spheres are connected by a massless non- conducting string of length d = 1.00 m, which is much greater than the radii of the spheres. (a) What is the electric potential energy of the system? (b) Suppose you cut the string. At that instant, what is the acceleration of each sphere? (c) A long time after you cut the string, what is the speed of each sphere as shown in figure 16.6 below.

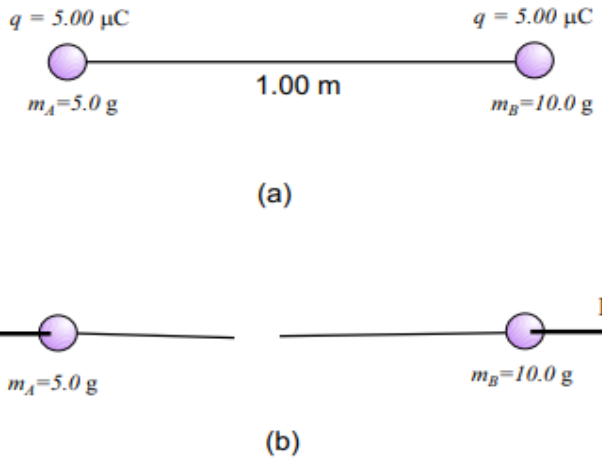


Fig 16.6

Review questions for lesson 16

16.1. A uniform electric field of magnitude 100 N/C is directed horizontally. A square sheet of side 40 cm is placed so that its plane makes an angle of 30° with the electric field. The electric flux through the sheet is:

A. 6.9 $N \cdot m^2/C$ B. 8.0 $N \cdot m^2/C$ C. 13.9 $N \cdot m^2/C$ D. 16.0 $N \cdot m^2/C$

16.2. Three points **P, Q**, and **R** are located at distances r, 2r, and 4r, respectively, from a positive point charge. Which statement is correct?

A. $V_P = V_Q = V_R$ C. $V_P > V_Q > V_R$ C. $V_R > V_Q > V_P$ D. $V_Q > V_P > V_R$

16.3. Two charges +2 μC and +3 μC are separated by 0.20 m. The electric potential energy of the system is:

A. 0.135 J B. 0.27 J C. 0.54 J D. 1.08 J

16.4. A uniform electric field has a magnitude of 250 V/m. Two points are separated by 0.40 m in the direction of the field. The potential difference between them is

A. 40 V B. 62.5 V C. 100 V D. 250 V

16.5. Which one of the following statements about equipotential surfaces is correct?

A. Electric field lines are parallel to equipotential surfaces.

B. Work is required to move a charge along an equipotential surface.

C. Equipotential surfaces can intersect each other.

D. Electric field lines are always perpendicular to equipotential surfaces.

Suggested Teaching Methods

- Use **vector diagrams** to distinguish the surface from the normal vector in flux calculations.
- **Gravitational Analogies:** Compare electrostatic potential energy with gravitational potential energy and a compressed spring.
- Demonstrate how electrostatic potential energy converts into kinetic energy using conservation of energy.

Teaching Resources

- **Core Material**: Grade 10- 11 Physics textbook units covering Electrostatics and Field Theory.
- **Props:** Flat sheets, grid maps, and arrows to explicitly showcase vector dot products.
- **Simulations:** PhET Charges and Fields simulation engine to map live visual grids of equipotential lines around multi-charge arrangements.

Expected Learning Outcomes: Students will successfully analyze spatial voltage distributions and perform structural calculations of flux and electric potential parameters. They will correctly evaluate energy conservation transformations during dynamic charge interactions without failing to separate potential configurations from localized field intensities.

Lesson 17(45 minutes)

Topic: Current Electricity

Sub-topics: Electric Current, Drift Velocity, Ohm's Law, and Series/Parallel Resistor Combinations

At the end of this lesson delivery, students will be able to:

- Define electric current, drift velocity, and current density with their respective SI units.
- Distinguish mathematically between conductivity, resistivity, and total resistance.
- State Ohm's Law and differentiate between ohmic and non-ohmic materials using I-V characteristic graphs.
- Identify the unique voltage, current, and resistance characteristics of series and parallel networks.
- Calculate the equivalent resistance, localized voltage drops, branch currents, and power distributions for complex network configurations.

17.1 Electric Current, Drift Velocity, and Current Density

Electric Current (I): The macroscopic rate at which net electric charge passes through a designated cross-sectional area of a conductor.

Electric current = $\frac{\text{charge}}{\text{time}} \rightarrow I = \frac{Q}{t}$ (1A = 1C/s)

Drift Velocity (V_d): The average microscopic velocity attained by free charge carriers (electrons) inside a conductor when influenced by an external electric field. The current is linked to this motion by:

$$\mathbf{I = ne\ A\ V_d} \quad \Rightarrow = \frac{I}{ne\ A}$$

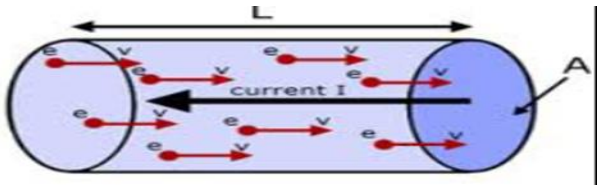


Figure 17.1 Motion of electrons through a conductor

Where **n** is charge carrier density, **A** is cross-sectional area, and **e** is elementary charge (1.6×10^{-19}C.

Current density(J)**:** The vector distribution of current flowing per unit cross-sectional area of a conductor.

$$\mathbf{J} = \frac{I}{A} = \text{ne } V_d \text{ (in A/m}^2\text{)}$$

Microscopic Ohm's Law*:* J = **σ** E, where **σ** is conductivity and E is electric field intensity.

Example 17.1 A current of 3.60 A flows through an automobile headlight. How many coulombs of charge flow through the headlight in a time of 2.60 hrs?

Solution: $I = \frac{Q}{t}$; Q = It = 3.37 x10^4 C

Example17.2 A copper wire with a diameter of 2mm carries a current of 15A. Calculate the drift velocity of electrons. (Given: Density of free electrons in copper, n = 8.5 x 10^{28} electrons /m^3, e = 1.6 x10^{-19} C).

Solution: $V_d = \frac{I}{ne\,A}$ = 3.5 x10^{-4} m/s

Exercise 17.1 The current supplied to a lamp with a 100-W light bulb is 0.87 amps. The lamp is wired using a copper wire with diameter 2.588 mm. Find the magnitude of the current density.

17.2 Resistivity and Resistance

Conductivity (σ) vs. Resistivity (ρ): Reciprocals of each other $\rho = \frac{1}{\sigma}$.

Conductivity: Measures how easily a material conducts electric current. Its units are Siemens per meter (**S/m**).

Resistivity: measures a material's intrinsic opposition to current flow (in **ohm-meter)** and depends exclusively on material composition and temperature.

Resistance (R): The geometric measurement of a specific object's opposition to current flow:

- Depends on resistivity **(ρ)**, length(L), temperature(T), and cross-sectional area(A).

$$\mathbf{R = \rho \frac{L}{A}}$$

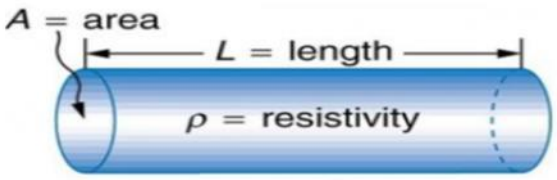


Fig 17.2 A uniform conductor of length L, and cross-sectional area A.

- R increases with **ρ** and **L**, and decreases with **A**.

Activity 17.1: How current density changes if a wire's diameter is doubled while maintaining a constant current.

Example: A 80.0 m Cu wire 1.0 mm in diameter is joined to a 49.0 m iron wire of the same diameter. The current in each is 2.0 A. a.) Find V in each wire.

Solution: $\boldsymbol{\rho}_{\mathrm{Cu}}$ = 1.7 x10^{-8} Ω. m, $\boldsymbol{\rho}_{\mathrm{Fe}}$ = 10 x10^{-8} Ω. m

$R_{Cu} = \rho \frac{L}{A} = 1.73\ \Omega$; $R_{Fe} = \rho \frac{L}{A} = 6.24\ \Omega$

$V_{Cu} = IR_{Cu} = 3.46V$, $V_{Fe} = IR_{Fe} = 12.48V$

Exercise 17.2 Calculate the current density, resistance, and electrical field of a 5-m length of copper wire with a diameter of 2.053 mm carrying a current of I = 10 mA. ($\boldsymbol{\rho}_{\mathrm{Cu}} = 1.7 \times 10^{-8}\ \Omega \cdot$ m).

17.3 Ohm's Law & Combinations of resistor

- **Ohm's Law:** At constant temperature and other physical conditions, current is directly proportional to voltage, I ∝ V (V = IR).
- Ohmic materials display a **linear I-V** relationship, while non-ohmic components exhibit **non-linear** behavior.

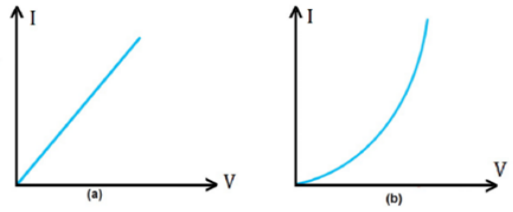


Figure 17.3 Current vs. voltage for an ohmic and non-ohmic material

Activity 17.2: Sketch two lines on the board: one perfectly linear through the origin (Ohmic) and one exponential curve representing a diode (Non-Ohmic). Have you turn to a neighbor and identify which graph permits a variable resistance value dependent on the operating voltage?

Series combination: is defined as when two or more resistors are connected end-to-end, they form a single path for the current flow.

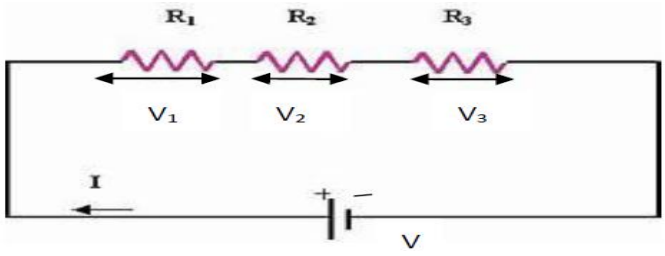


Fig 17.4 Fig17.5 For a series connection of three resistors.

- Key Characteristics of series connection:

1. The same current (I) passes through all the resistances in a circuit.

$I = I_1 = I_2 = I_3 = \ldots = I_n$

2. The potential difference is different across each resistance i.e. Total (net) voltage across the combination is the **sum** of voltage drops across each resistor:

$V_{total} = V_1 + V_2 + V_3 + \ldots\ldots$

3. The total or equivalent resistance $\mathbf{R_{eq}}$ is equal to the **sum** of all individual resistances:

$R_{eq} = R_1 + R_2 + R_3 + \ldots\ldots$

4. $I_{total} = \frac{V_{total}}{R_{eq}}$

Parallel Connection: is an arrangement where two or more resistors are connected across the same two points, so that the same voltage acts on each resistor.

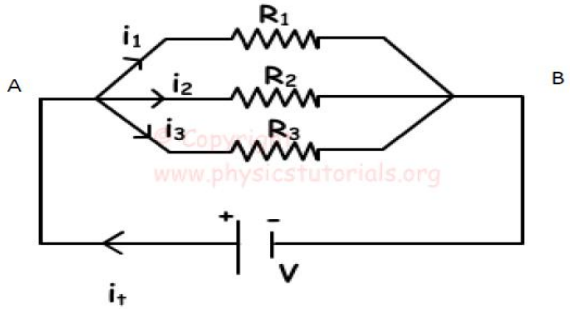


Fig17.5 For a parallel connection of three resistors.

- Key Characteristics of parallel connection:

1. The voltage or potential difference across each resistance is the same.

$V = V_1 = V_2 = V_3 =, \ldots\ldots$

2. The current passing through each resistance is different. The total current among the resistors is the **sum** of current flow through each resistor:

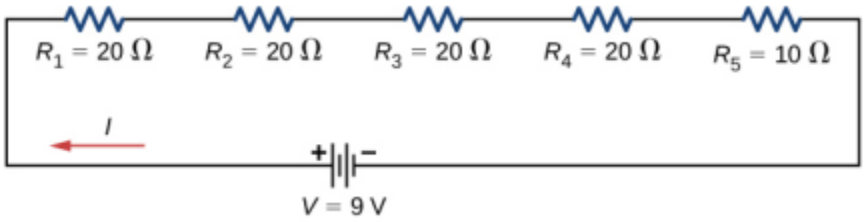
$I_{total} = I_1 + I_2 + I_3 + \ldots\ldots$

3. The total or equivalent resistance **($\mathbf{R_{eq}}$)** is equal to the reciprocal **sum** of all individual resistances:

$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \ldots\ldots$

4. $I_{total} = \frac{V_{total}}{R_{eq}}$

Example 17.3 A battery with a terminal voltage of 9 V is connected to a circuit consisting of four 20-Ω and one 10-Ω resistors all in series (Figure 17.6). Assume the battery has negligible internal resistance. (a) Calculate the equivalent resistance of the circuit. (b) Calculate the current through each resistor. (c) Calculate the potential drop across each resistor. (d) Determine the total power dissipated by the resistors and the power supplied by the battery.



Fir 17.6

Solution:

a. $R_{eq} = 90\ \Omega$,

b. $I_{total} = \frac{V_{total}}{R_{eq}} = 0.1A$

c. $V_1 = V_2 = V_3 = V_4 = I_{total} \text{x } 20\ \Omega = 2V$; $V_5 = I_{total} \text{x } 10\ \Omega = 1V$

d. The power dissipated by a resistor is equal to $P = I^2 R$, and the power supplied by the battery is equal to $P = I_{total}\ V_{source}$.

$P_1 = P_2 = P_3 = P_4 = (0.1\ A)\ 2\ (20\ \Omega) = 0.2\ W$,

$P_5 = (0.1\ A)\ 2\ (10\ \Omega) = 0.1\ W$,

$P_{Dissipated} = 0.2\ W + 0.2\ W + 0.2\ W + 0.2\ W + 0.1\ W = 0.9\ W$,

$P_{source} = I_{total}\ V_{source} = (0.1\ A)(9\ V) = 0.9\ W$.

Exercise 17.3. Figure 17.7 shows resistors wired in a combination of series and parallel. We can consider R1 to be the resistance of wires leading to R_2 and R_3. (a) Find the equivalent resistance of the circuit. (b) What is the potential drop V_1 across resistor R_1? (c) Find the current I_2 through resistor R_2. (d) What power is dissipated by R_2?

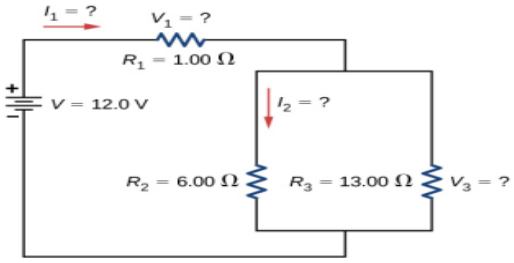


Fig17.7

Review questions for lesson 17

17.1. Current density is greatest in a conductor when:

A. the current is small and the cross-sectional area is large

B. the current is large and the cross-sectional area is small

C. both current and area are large

D. both current and area are small

17.2. If the number density of free electrons is doubled while the drift velocity remains constant, the current density will be:

A. remain unchanged C. become half

B. Double D. become four times

17.3. In a metal conductor, the direction of conventional current is

A. the same as the direction of electron drift

B. opposite to the direction of electron drift

C. perpendicular to the direction of electron drift

D. independent of electron motion

17.4. A copper wire has a diameter of 2.0 mm and carries a current of 10 A. If the diameter of the wire is doubled while the current remains unchanged, the drift velocity of electrons becomes

A. Four times larger B. Twice as large C. Half as large D. One-fourth as large

17.5. Two wires are made of the same material. Wire X has twice the length and half the diameter of wire Y. The resistance of wire X compared with wire Y is

A. $2R_Y$ B. $4R_Y$ C. $8R_Y$ D. $16R_Y$

17.6. A resistor obeys Ohm's law. If the voltage across it is increased from 6 V to 18 V, while its temperature remains constant, the current will

A. remain unchanged C. become twice its original value

B. become three times its original value D. become six times its original value

17.7. A combination of a 6 Ω and a 3 Ω resistor is connected in series, and this combination is in parallel with a 2 Ω resistor across 12 V. what is the total current drawn from battery:

A. 6 A C. 7 A

B. 10 A D. 12 A

17.8. Determine the current in the circuit as shown in figure 17.8 below.

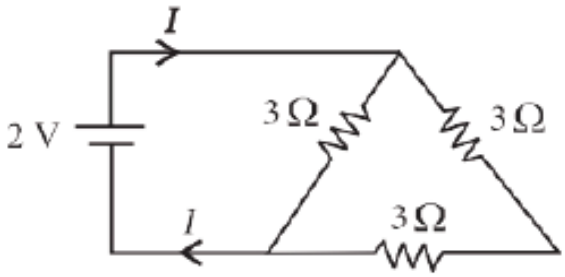


Fig 17.8

Suggested Teaching Methods

- **Analogy-Based Instruction:** Compare electric current and resistance to fluid dynamics (water flow through pipes of varying diameters) to establish physical intuition.
- **Circuit Reduction Scaffold Method:** Visually enclose parallel sub-sections in boxes on the board, replacing them step-by-step with single equivalent values.
- **Active Peer Verification:** Have students complete Exercise 17.3 independently, then cross-verify step values with adjacent peers before revealing the final step calculations.

Teaching Resources

- **Standard Materials**: Grade 9-12 Physics Student Textbooks and official classroom guidelines.
- **Laboratory Hardware**: Constant DC power banks, varied ohmic carbon resistors, digital multimeters, and copper connecting leads.
- **Digital Visualization**: PhET Circuit Construction Kit (DC) to demonstrate current splitting at parallel branch points.7

Expected Learning Outcomes

- Students will be able to structurally break down a mixed combination circuit network into its basic series and parallel components.
- They will display mathematical proficiency in translating macroscopic circuit constraints (V, I, R) into microscopic conductor behavior dynamics (J, σ, V_d) without formula confusion.

Lesson 18(45 minutes)

Topic: Current Electricity

Sub-topics: Electrical Measuring Instruments and the Wheatstone Bridge

At the end of this lesson delivery, students will be able to:

- Explain the operational differences between an ammeter and a voltmeter, including how they are connected to a circuit.
- Calculate the required shunt resistance (R_{Sh}) to convert a galvanometer into an ammeter.
- Calculate the required multiplier resistance (R_m) to convert a galvanometer into a voltmeter.
- Analyze potential divider circuits to find output voltages under variable conditions.
- Describe the operational mechanisms and benefits of null-method instruments (potentiometers and Wheatstone bridges).
- Solve balance equations to determine unknown resistances using a Wheatstone bridge network.

18.1 Ammeter vs. Voltmeter Conversion

A standard galvanometer is **highly sensitive** and can only handle small currents (I_g). To measure larger electrical quantities, it must be modified.

- **Ammeter Conversion:** Used to measure large currents and must be connected in series within a circuit. An ideal ammeter has zero internal resistance.
- **Modification***:* A low-resistance resistor called a **shunt** (R_{Sh}) is connected in **parallel** with the galvanometer.
- **Formula***:* Because voltage across parallel branches is equal:

$V_g = V_{Sh} \rightarrow I_g R_g = I_{Sh} R_{Sh}$ since $I_{Sh} = I - I_g$

$$\mathbf{R_{Sh} = \left(\frac{I_g}{I - I_g}\right) R_g}$$

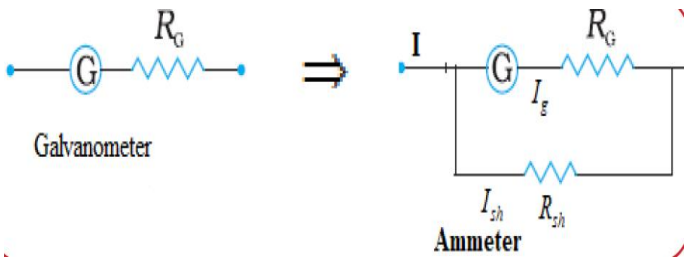


Fig 18.1 Converting galvanometer to an ammeter

- **Voltmeter Conversion:** Used to measure potential differences and must be connected in parallel across a component. An ideal voltmeter **has infinite internal** resistance.
- **Modification:** A high-resistance resistor called a **multiplier (R_m)** is connected in **series** with the galvanometer.
- **Formula*:*** The total voltage drop is the sum of individual series drops:

$V = V_g + V_M \rightarrow I_g R_g + I_g R_M \quad \rightarrow \quad I_g(R_g + R_M)$

$R_g + R_M = \frac{V}{I_g} \rightarrow \quad \mathbf{R_M = \frac{V}{I_g} - R_g}$

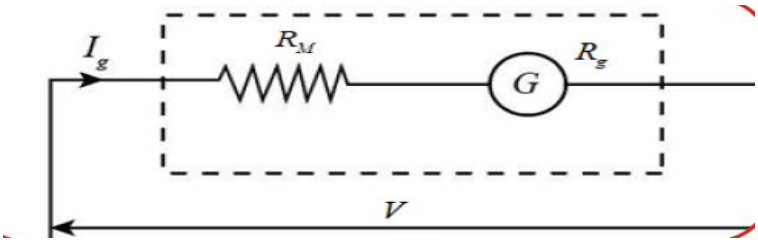


Fig 18.2

Where R_M = multiplier resistance, $\mathbf{R_g}$ = the resistance of a galvanometer, V = the potential difference to be measured by the voltmeter, and I_g = galvanometer current.

Example 18.1: A galvanometer of full-scale deflection 5 mA is to be converted into a 10 A ammeter. If its coil has a resistance of 50 Ω, what value shunt must be fitted?

Solution: I = 10A; I_g = 5mA; R_g = 50 Ω; R_{Sh} =?

$R_{Sh} = \left(\frac{I_g}{I - I_g}\right) R_g \quad = 0.025\ \Omega$

Example18.2: A galvanometer has a resistance of 40 Ω and is of 3 mA full scale deflection. How would you modify it to a 0-10 V voltmeter?

Solution: V = 10V; R_g = 40 Ω; I_g = I = 3 mA; R_M=?

$R_M = \frac{V}{I_g} - R_g = \quad 3333.33\Omega - 40\ \Omega = 3.29k\ \Omega$

Exercise 18.1: A galvanometer has full-scale deflection Ig =2 mA, and coil resistance R_g =100 Ω. Convert it into a **5 A.** What shunt is required?

Exercise 18.2: A galvanometer has a resistance of 50 Ω and gives full-scale deflection at 2 mA. How would you convert it into a 0-5 V voltmeter?

18.2 Potential Divider, Potentiometer and Wheatstone Bridge

Potential Divider: A simple series resistor network that yields a fraction of the input voltage.

- The output voltage drops across a target resistor R_2 (**connected load)** is governed by:

$$\mathbf{V_{out} = V_{in}\ \frac{R_2}{R_{1+R_2}}} \qquad \textbf{since } \mathbf{I_{total} = \frac{V_{in}}{R_{1+R_2}}}$$

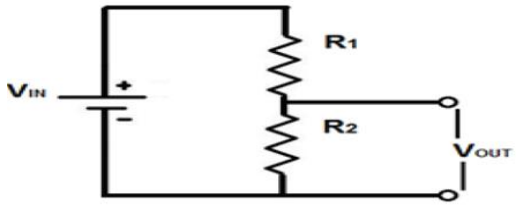


Fig 18.3 Potential divider circuit

- **Drawbacks***:* It draws current from the source, meaning its output drops significantly if a load is connected, leading to power loss.
- Applications: sensor circuits, transistor biasing, adjustable voltage supply.

Potentiometer: A precision instrument used to measure an unknown electromotive force (emf) using a null-method (**zero current deflection**).

- Because it **draws no current** from the cell being tested at the balance point, it eliminates errors caused by the cell's internal resistance, making it superior to a standard voltmeter.

Wheatstone Bridge: A structural network of four resistors arranged in a loop used to determine an **unknown resistance**. When the variable arms are adjusted such that the current passing through the central galvanometer is exactly zero ($I_g = 0$), the bridge is **balanced**.

➢ Under this balanced Condition

$V_{AC} = V_{AD} \rightarrow I_1R_1 = I_2R_2$

And also: $V_{CB} = V_{DB} \rightarrow I_3R_3 = I_4R_4$

Where I_1, I_2, I_3, and I_4 are current in resistance arms respectively, since $I_1 = I_3$, $I_2 = I_4$.

$$\frac{\mathbf{R_1}}{\mathbf{R_3}} = \frac{\mathbf{R_2}}{\mathbf{R_4}}$$

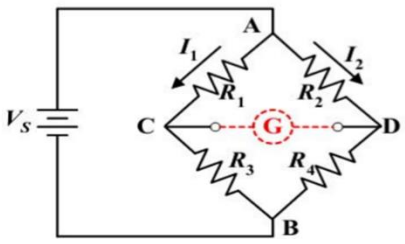


Fig 18.4 A Wheatstone bridge

Example 18.3: A circuit consists of two resistors with resistances $R_1 = 6.0\ \Omega$ and $R_2 = 1.5\ \Omega$, a variable resistor, the resistance R_{var} of which can be adjusted, a resistor of unknown value Ru, and 9.0-volt battery connected as shown in the figure. When $\mathbf{R_{var}}$ is adjusted to 12Ω, there is zero current through the ammeter. What is the unknown resistance $\mathbf{R_u}$?

Solution: Wheatstone bridge is balanced, so:

$$\frac{R_1}{R_{var}} = \frac{R_2}{R_u}$$

$$R_4 = R_{var}\frac{R_2}{R_1} \rightarrow \frac{1.6\ x\ 12}{6} = 3\Omega$$

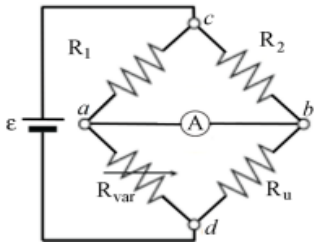


Fig 18.6

Exercise 18.3 A Wheatstone bridge consists of four resistors as shown figure 18.7below. The value of S = x must be:

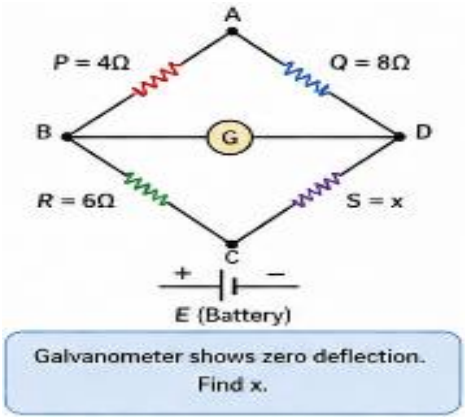


Fig 18.7

Activity 18.1: You have a galvanometer. If you accidentally connect a low-resistance shunt in series instead of parallel, or a high-resistance multiplier in parallel instead of series, what will happen to the device when you turn on the power? Discussing in groups?

Review questions for lesson 18

18.1. A galvanometer is converted into an ammeter by connecting a low-resistance shunt in parallel. The primary purpose of the shunt resistor is to:

A. increase the sensitivity of the galvanometer.

B. allow most of the current to bypass the galvanometer coil.

C. increase the resistance of the instrument.

D. make the galvanometer measure voltage.

18.2. A galvanometer is converted into a voltmeter by connecting a large resistance in series because the added resistance:

A. allows a larger current through the galvanometer.

B. reduces the current through the galvanometer while allowing measurement of larger voltages.

C. decreases the total resistance of the instrument.

D. prevents any current from flowing through the galvanometer.

18.3. Why is a potentiometer considered superior to a standard digital voltmeter for measuring the electromotive force (EMF) of a cell?

A. It draws no current at balance

B. It operates on the heating effect

C. It has a lower internal resistance

D. It is easier to portable use

18.4. Which one of the following statements is **NOT** correct?

A. An ideal ammeter has very low internal resistance and is connected in series.

B. An ideal voltmeter has very high internal resistance and is connected in parallel.

C. A galvanometer is converted into a voltmeter by connecting a high resistance in parallel with it.

D. A galvanometer is converted into an ammeter by connecting a low-resistance shunt in parallel with it.

18.5. In a balanced Wheatstone bridge, the galvanometer reads zero because:

A. the galvanometer has infinite resistance.

B. equal currents flow through the galvanometer in opposite directions.

C. the two points connected by the galvanometer are at the same electric potential.

D. no current flows in the bridge circuit.

18.6. A galvanometer has a full-scale deflection current of 5 mA and an internal resistance of 50 Ω. It is converted into a 10 A ammeter by connecting a shunt resistor. The value of the shunt resistor is closest to:

A. 0.025Ω
C. 25Ω
B. 100Ω
D. 0.25Ω

18.7. A potential divider consists of two resistors $R_1 = 200$ Ω and $R_2 = 600$ Ω connected in series across a 12 V DC supply. What is the output voltage across the 600Ω resistor when no external load is connected?

A. 3V
C. 9V
B. 6V
D. 12V

Suggested Teaching Methods

- **Direct Instruction / Mathematical Derivation:** Explicitly model the algebraic substitutions that transform basic Ohm's Law variations into specialized instrument conversion rules.
- **Structured Problem-Solving Scaffolding:** Guide the class step-by-step from analyzing a verbal scenario description to setting up cross-multiplication parameters for bridge networks.
- **Comparative Group Inquiry:** Contrast the active current draw of potential dividers against the zero-draw characteristics of null-balance instruments.

Teaching Resources

- Textbooks: High School Physics Grade 9-12 Student Text and Teacher Support Materials.
- Apparatus / Laboratory Items: Unmodified moving-coil galvanometers, resistance boxes, connecting wires, slide-wire potentiometers, and DC power packs.
- Digital Material: PhET circuit building web modules for observing structural current splitting paths.

Expected Learning Outcomes

Students will possess the technical competence to select appropriate shunt or series resistors to modify meter sensitivities. They will also be capable of evaluating balancing criteria in bridge topologies to calculate unknown resistive component characteristics without error.

Lesson 19(45 minutes)

Topic: Current Electricity

Sub-topics: Kirchhoff's (Junction and Loop) Rules) and Bulb Brightness

At the end of this lesson delivery, students will be able to:

- State and explain Kirchhoff's Junction Rule based on the law of conservation of charge.
- State and explain Kirchhoff's Loop Rule based on the law of conservation of energy.
- Calculate unknown currents and voltages across components in a complex circuit using Kirchhoff's rules.
- Analyze and predict the relative brightness of lightbulbs in series, parallel, and bridge network configurations based on electrical power calculations

19.1 Kirchhoff's Rules

Kirchhoff's Rules: are used to analyze complex electrical circuits where series and parallel combinations occur. There are two rules:

Kirchhoff's first rule (Junction rule): At any junction in an electrical circuit, the total current entering a junction **must be equal** to the total current leaving the junction.

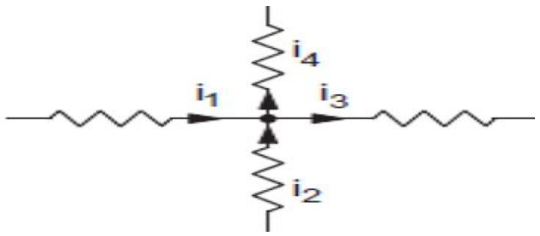


Figure 19.1 Kirchhoff's junction rule

$\sum \mathbf{I_{in}} = \sum \mathbf{I_{out}} \rightarrow \mathbf{I_1 + I_2 = I_3 + I_4}$ **(**Based on **conservation of charge**).

- **Sign Convention***:* Currents entering a junction are treated as positive (+); currents leaving a junction are treated as negative (-).

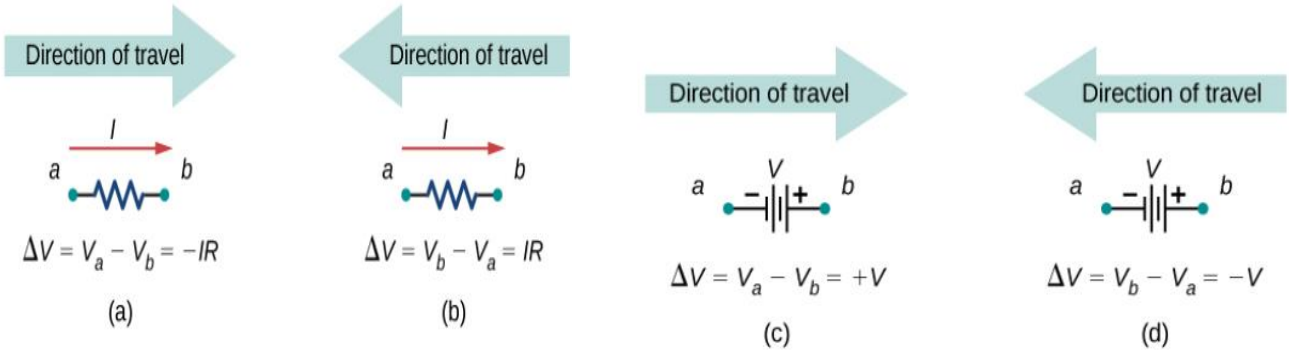
Kirchhoff's Second Rule (Loop Rule): States that the **algebraic sum** of changes in potential around any closed circuit path (loop) must be zero:

$\sum \boldsymbol{V} = \mathbf{0}$ **(**Based on **conservation of energy).**

➢ Review sign conventions carefully:

- Moving with conventional current across a resistor is $\mathbf{V_{across\ a\ resistor} = -IR}$; Moving against current is $\mathbf{V_{across\ a\ resistor} = +IR}$

- Moving from a sources of EMF (Cells): $\Delta V = +\boldsymbol{\varepsilon}$ if traversing from the **– to +** terminal; $\Delta V = -\boldsymbol{\varepsilon}$ if traversing from the **+ to -** terminal.



Activity 19.1: Sketch a complex node on the board with three incoming currents (3A, 4A, and an unknown I_x) and two outgoing currents (2A and 6A). Have students turn to a partner, state the conservation law that dictates the balance, and calculate I_x.

Activity 19. 2: Two bulbs, Bulb A (R = 10 ohm) and Bulb B (R = 20 ohm), are connected first in series and then in parallel to an ideal source.

a. Ask students to write down an individual prediction of which bulb shines brighter in each arrangement.

b. Guide a brief discussion emphasizing why the "higher resistance = brighter" rule reverses when switching from series to parallel.

19.1 Bulb Brightness

The operational **brightness** of an incandescent lightbulb is directly proportional to the actual electric power (P) it dissipates, measured in lumens (light output unit). Power depends on **current through** the bulb and its **resistance**:

- Higher power → brighter bulb it shines, while lower power → dimmer bulb

Connection	**Voltage/Current**	**Power**	**Brightest Bulb**
Series	Same current	$\mathbf{P = I^2R}$	Higher resistance brighter
Parallel	Same voltage	$\mathbf{P = \frac{V^2}{R}}$	Lower resistance brighter
Balanced Wheatstone	No current in middle	0	Middle bulb off
Unbalanced Wheatstone	Current divides	$\mathbf{P = I^2R}$	Depends on series-parallel division

Example 19.1 This diagram shows a part of a circuit**.** What are the currents at X and at Y, and show their directions shown figure 19.2 below?

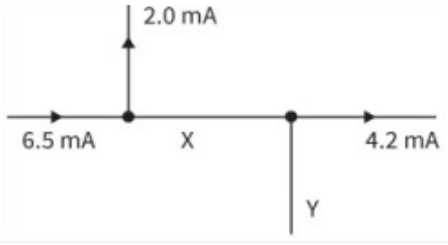


Fig 19.2

Solution: We solve this using **Kirchhoff's Current Law (KCL)**:

Step 1: Analyze the first junction **(Current X),** from the diagram:

- **6.5 mA** enters the first junction from the left.
- **2.0 mA** leaves upward.
- The remaining current continues to the right as **X**.

Current entering = Current leaving

6.5mA = 2mA+X; X = 4.5mA (So X is 4.5 mA to the right).

Step 2: Analyze the second junction **(Current Y)**

- **4.5mA** enters from the left (current X).
- **4.2 mA** leaves to the right.
- The remaining current must flow downward through **Y**.

4.5 = 4.2+Y

Y= 4.5- 4.2 = 0.3mA (So Y = 0.3 mA downward).

Example 19.3 What is the current I_1 in this circuit diagram 19.3 below? (Hint: Left junction point = **A**; Right junction point = **B)**

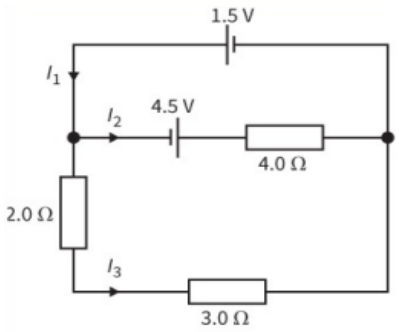


Fig 19.3

Solution: Using Kirchhoff's Rule

- Apply Kirchhoff's Current Law: $I_1 = I_2+I_3$

Branch1(top): 1.5 V battery only, Positive terminal is on the left, so

- Thus the potential difference across every branch is $V_{AB} = V_A-V_B = +1.5V$

Branch 2 (Middle):

- Contains 4.5 V battery and 4 Ω resistor
- Current is I_2 from left to right.

- Applying KVL: **loop1(clockwise direction)**

1.5 V- 4.5V = $+4I_2$

I_2 = -0.75A (The negative sign means the actual current is **0.75 A** form **right to left).**

- Applying KVL: **loop2(clockwise direction)**

1.5 V = $2I_3 +3I_3$

I_3 = 0.3A

Therefore, $I_1 = I_2+I_3$ = -0.45A (The negative sign means the actual current is **opposite the indicated arrow**).

Exercise 19.1. Find all the currents and voltages across each resistor and cell in the figure 19.4below.

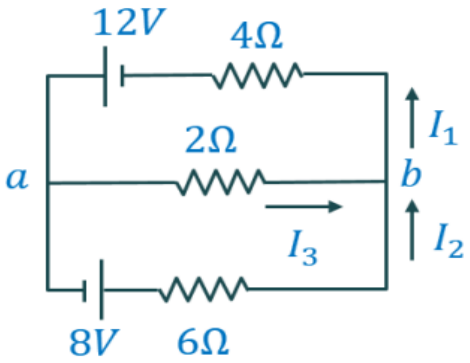


Fig 19.4

Review questions for lesson 19

19.1. A student applies Kirchhoff's Loop Rule while moving clockwise around a closed circuit. Which statement is **correct**?

A. Crossing a resistor in the direction of current gives a potential rise of +IR.

B. Crossing a battery from its positive to negative terminal gives a potential rise.

C. Crossing a resistor opposite to the current gives a potential rise of +IR.

D. The algebraic sum of currents around a closed loop is zero.

19.2. Three identical bulbs are connected in different circuits.

Circuit I: Three bulbs connected in **series**.

Circuit II: Three bulbs connected in **parallel**.

Circuit III: A balanced Wheatstone bridge with one bulb in the middle branch.

Which statement is correct?

A. Bulbs in Circuit I are brightest because they carry the largest current.

B. Bulbs in Circuit II are brightest because each receives the full battery voltage.

C. The middle bulb in Circuit III is brightest because current passes through all branches.

D. All bulbs have equal brightness in all three circuits.

19.3. Kirchhoff's Junction Rule is a direct consequence of the law of conservation of:

A. Energy B. Momentum C. Charge D. Mass

19.4. Two bulbs have resistances: $R_1 = 4\Omega$, $R_2 = 8\Omega$. They are connected in series; which bulb is brighter?

A. 8 Ω bulb B. 4 Ω bulb C. Equal brightness D. Cannot be determined

19.5. At a junction in an electrical circuit, three currents are directed toward the junction with magnitudes 2 A, 3 A, and I. Two currents of 4 A and 5 A are directed away from the junction. The value of I is:

A. 2A B. 3A C. 4A D. 6A

Suggested Teaching Methods

- **Interactive Lecture / Scaffolded Problem Solving:** Introduce the core tenets of KCL and KVL, explicitly mapping out the sign convention matrix on the board.
- **Think-Pair-Share:** Used during conceptual checkpoints regarding bulb power dynamics to surface and correct common student misconceptions.

- **Step-by-Step Explicit Modeling:** Working through linear multi-loop equations methodically on the board to build structural mathematical confidence.

Teaching Resources

- **Textbooks & Guides**: Physics Grade 9-12 Student Textbook and Teacher's Guide.
- **Digital Aids**: PhET Interactive Simulations (Circuit Construction Kit: DC).
- **Visual Aids**: Chalkboard diagram tracking color-coded loop orientations and junction current pathways.

Expected Learning Outcomes

Students will demonstrate the ability to construct loop and junction algebraic equations from given physical circuit schematics. They will produce accurate numerical evaluations of branch currents and component voltage drops, and mathematically justify why specific lighting components glow with differing relative intensities based on circuit placement.

Weekly Assessment & Revision for lesson 16-19 (Every Friday)

Every Friday should be dedicated to **assessment and revision** of the concepts and skills covered during the week.

The primary purpose of the weekly assessment is to:

- Monitor students' attendance and participation.
- Evaluate students' understanding of the week's lessons.
- Identify learning gaps and misconceptions.
- Provide timely feedback and make instructional adjustments where necessary.

Prepare **25–30 assessment questions** using a variety of item types (e.g., true/false, multiple choice, short-answer, problem-solving or conceptual questions) as appropriate.

Suggested Time Allocation for Friday Assessment

- **30 minutes – Assessment:** Students complete the assessment independently under examination conditions.
- **15 minutes – Review and Feedback:** The teacher reviews the answers with the class, explains the correct solutions, addresses common misconceptions and provides constructive feedback.

Lesson 21(45 minutes)

Topic: Current Electricity

Sub-topic: Capacitors, Capacitance, & Dielectrics

At the end of this lesson delivery, students will be able to:

- Define capacitance, explain the structural properties of a parallel-plate capacitor,
- Explain the effect of inserting dielectric in the gap between the plates of a capacitor.
- Solve problems related to capacitance of parallel plate capacitor.

21.1 Capacitance and Capacitance

- **Capacitor:** An electrical component designed to store electric charge and potential energy within an internal electric field.
- It typically consists of two isolated parallel conducting plates separated by an insulating medium (vacuum, air, or a dielectric material).
- **Capacitance (C):** The quantitative measure of a capacitor's ability to store charge per unit of potential difference across its plates:

$$\mathbf{C} = \frac{Q}{V}$$



Fig 21.1 Capacitor, with the positive plate carrying a positive charge and the negative plate an equal amount of negative charges.

- **SI Unit:** The Farad (F), where 1 F = 1 C/V.
- Because 1F represents an exceptionally large volume of charge, fractional units are preferred in real-world applications: 1 μF = 10^{-6}F, 1nF = 10^{-9}F, and 1pF = 10^{-12}F.

21.2 The parallel-plate Capacitor

- A standard parallel-plate capacitor features two matching conducting planes of surface area **A** separated by a small gap distance **d.**

➢ Capacitance in Air/Vacuum (C_o):

$$C = \varepsilon_o \frac{A}{d}$$

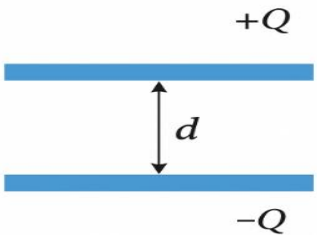


Figure 21.2 Parallel-plate Capacitor

Where: A = plate area, d = separation between plates, ε_o = 8.85×10^{-12} F/m (permittivity of free space).

- Increasing plate area (A) provides more surface space to accumulate charge, increasing capacitance.
- Decreasing separation distance (d) maximizes the electrostatic attraction between opposite plates, strengthening the field and increasing capacitance.

Internal Electric Field (E): The field between the plates is uniform, directed away from the positive boundary and toward the negative boundary:

$$\mathbf{E} = \frac{\boldsymbol{\sigma}}{\boldsymbol{\varepsilon_0}} = \frac{\boldsymbol{Q}}{\boldsymbol{\varepsilon_o A}}$$

Where the charge density $\boldsymbol{\sigma}$ on each of these surfaces has a magnitude:

$$\boldsymbol{\sigma} = \frac{\boldsymbol{Q}}{\boldsymbol{A}}$$

➢ The potential difference between each plate is given by:

$$V = Ed = \frac{\boldsymbol{Q\,d}}{\boldsymbol{\varepsilon_o A}}$$

Dielectrics: is an insulating material placed between capacitor plates. Examples: glass, mica, plastic, paper, air, etc.

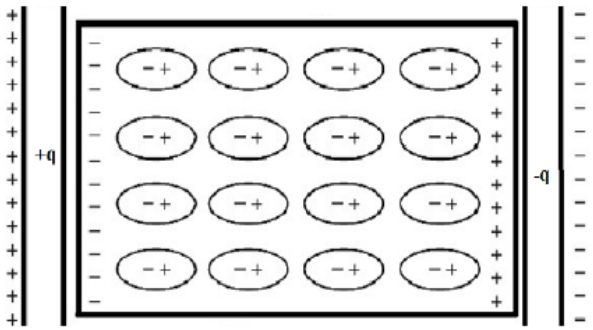


Fig 21.3 Polarization in a dielectric

Note that: Let dielectric constant = k (>1)

- **Case A: Battery remains connected (V = constant)**
  - Capacitance increases, charge increases, electric field decreases and energy increases.
- **Case B: Battery disconnected (Q = constant)**
  - Capacitance increases, voltage decreases, electric field decreases, and energy decreases.

Activity 21.1: Predict how the total capacitance changes if the area of a parallel-plate capacitor is tripled while its separation distance is halved.

Example 21.1: The capacitance of a parallel-plate capacitor with air as the medium is 6 μF. When a dielectric material is inserted between the plates, the capacitance increases to 30 μF. Determine the permittivity of the dielectric medium. (Use ε_o = 8.85×10^{-12} F/m permittivity of free space).

Solution: C = C_o k , k = $\frac{C}{C_o}$ = 5

Exercise 21.1: A parallel-plate capacitor has oil between its plates, with dielectric constant k = $\sqrt{2}$. The capacitance with the oil in place is **C**. If the oil is removed, what is the new capacitance of the capacitor?

Review Questions for lesson 21

21.1. A parallel-plate capacitor is connected to a battery. Which of the following changes will increase its capacitance?

A. Doubling the plate separation

B. Halving the plate area

C. Doubling the plate area while keeping separation constant

D. Increasing the battery voltage

21.2. A parallel-plate capacitor has plate area A and separation d. Its capacitance becomes six times its original value if:

A. Area is doubled and separation tripled

B. Area is tripled and separation halved

C. Area is tripled and separation reduced to one-half while a dielectric of k =1 is inserted

D. Area is tripled and separation reduced to half

21.3. A capacitor remains connected to a battery while a dielectric of dielectric constant k is inserted. Which statement is correct?

A. Charge remains constant because voltage is fixed.

B. Charge increases because capacitance increases.

C. Voltage increases.

D. Stored energy decreases.

21.4. A 4μ capacitor is connected in series with a 6μF capacitor across a 12 V battery. The charge on the 6μ capacitor is:

A. 12 μC B. 24 μC C. 48 μC D. 6 μC

21.5. The capacitance of a parallel-plate capacitor is C. If the area is doubled and the separation is halved, the new capacitance is:

A. C/4 B. 4C C. 2C D. ½ C

21.6. An air-filled parallel-plate capacitor is to be constructed to store 12 μC of charge when operated at 1200 V. What are the minimum plate separation and the minimum plate area required? (Electric field strength of air = 3×10^6V/m).

Suggested Teaching Methods

- **Visual Modeling:** Use simple diagrams to show how charges line up on the plates and how dipoles align inside a dielectric material to illustrate polarization.
- **Variable Extraction Scaffolding:** Help students practice identifying whether a problem states the battery is connected (V) is constant) or disconnected (Q) is constant) before they begin any math.
- **Interactive Socratic Questioning:** Guide the class through the relationship between area, gap distance, and total capacitance using step-by-step questions.

Teaching Resources

- **Textbooks:** High School Physics Grade 9-12 Reference Guides.
- **Visual Aids:** Two flat metal sheets and a book to act as a physical model of a parallel-plate capacitor with a dielectric.
- **Simulations:** PhET *Capacitor Lab* simulation to show real-time changes in charge, field lines, and capacitance when altering geometry or adding dielectrics.

Expected Learning Outcomes

Students will be able to explain the physical characteristics of capacitance and dielectrics. They will also be able to solve quantitative problems involving parallel-plate dimensions and calculate changes in charge, voltage, or energy when a dielectric is added.

Lesson 22(45 minutes)

Topic: Current Electricity

Sub-topic: Capacitor Combinations

At the end of this lesson delivery, students will be able to:

- Identify series and parallel capacitor configurations within schematic circuit diagrams.
- Derive and apply equivalent capacitance formulas for both series and parallel connections.
- Calculate total charge, localized branch charges, and voltage drops across individual components in a multi-capacitor circuit.

22.1 Combination of Capacitors

Capacitors can be connected in series, parallel, or in mixed networks. Their effective capacitance depends on how charges and voltages distribute in the circuit.

Capacitors in Series:

- Capacitors are in series when they are connected end-to-end in a single path.
- The same charge (Q) flows and stays on each capacitor.
- The total voltage is the sum of voltages across each capacitor.
- The equivalent capacitance becomes smaller than any individual capacitor.

1. $Q_s = Q_1 = Q_2 = Q_3$,
2. $V_s = V_1 + V_2 + V_3$,
3. $C_{eq} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}$

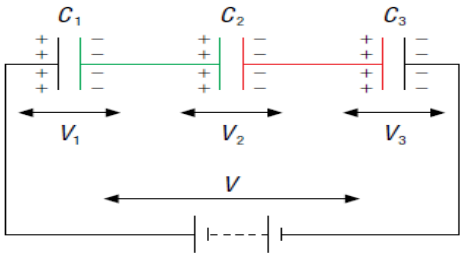


Figure 22.1 Three capacitors in series. Each has the same charge

Capacitors in Parallel

- Capacitors are in parallel when both their plates are connected to the same two points.
- The **same voltage (V)** appears across all capacitors.
- The **total charge** is the sum of charges on each capacitor.
- The equivalent capacitance becomes **larger** than any individual capacitor

1. $V_s = V_1 = V_2 = V_3$,
2. $Q_T = Q_1 + Q_2 + Q_3$
3. $C_{eq} = C_1 + C_2 + C_3$

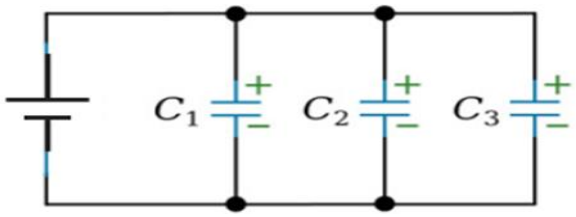


Fig 22.2 Three capacitors in parallel each has the same charge

Activity 22.1 Two capacitors (C_1 = 2μF and C_2 = 100 μF) are wired in series. Identify which component stores a larger volume of charge and explain their reasoning.

Example22.1: A network of four capacitors is connected to a 22 V voltage source, as shown in Figure 22.3 below. The capacitances are C_1=3 μF, C_2= 6 μF, C_3 = 9 μF, and C_4=12 μFC_. Determine the equivalent capacitance of the network and the total charge stored.

Solution:

- The capacitors C_1, C_2 & C_3 are in series: $C_{123} = \frac{18}{11}$ μF
- The capacitor C_{123} and C_4 are in parallel: $C_{1234} = \frac{150}{11}$ μF
- $Q_T = C_{1234}$ x $V_s = \frac{150}{11}$ μF x 22V= 300 μC

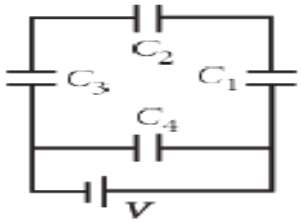


Fig 22.3

Exercise 22.1 In the circuit shown in Figure 22.4 below. What is the charge on the 2 μF capacitor?

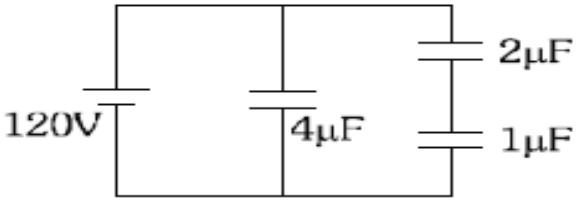


Fig 22.4

Review Questions for lesson 22

22.1. Which of the following statements correctly compares capacitors connected in series and in parallel?

A. In series, voltage is the same across each capacitor; in parallel, charge is the same.

B. In both series and parallel, each capacitor stores the same charge.

C. In series, each capacitor has the same charge; in parallel, each capacitor has the same voltage.

D. Equivalent capacitance is always smaller in parallel than in series.

22.2. Two capacitors, of 2.0 and 4.0 μF capacitance, are connected in parallel across a 300 V potential difference as shown figure 22.5 below. Calculate the charge stored on each capacitor.

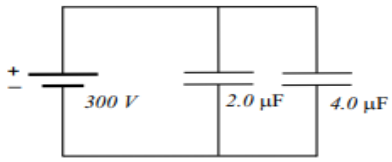


Fig 22.5

22.3. Find the equivalent capacitor and the total amount of charge on capacitor shown in figure 22.6 below.

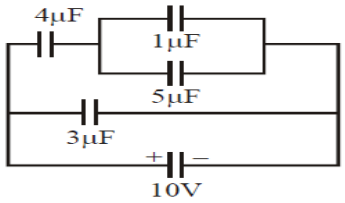


Fig 22.6

Suggested Teaching Methods

- **Step-by-Step Circuit Reduction:** Teach students to simplify complex circuit networks systematically by replacing groups of parallel or series components with their calculated equivalent values.
- **Guided Problem-Solving:** Work through example 22.1 on the board step-by-step to show how to split calculations between series and parallel sections.

Teaching Resources

- **Textbooks:** High School Physics Grade 9-12 Reference Materials.

Expected Learning Outcomes

Students will be able to calculate equivalent capacitance, individual component voltages, and charge distribution profiles for mixed capacitor networks.

Lesson 23(45 minutes)

Topic: Simple machine

Sub-topic: Machine Parameters (Load, Effort, Work Input/Output), Mechanical Advantage, Velocity Ratio, Efficiency, and Functional Purposes

At the end of this lesson delivery, students will be able to:

- Define and calculate basic machine parameters (Load, Effort, Work Input, Work Output, Mechanical Advantage, Velocity Ratio, Efficiency)
- Classify simple machines based on their primary functions as force multipliers, speed/distance multipliers, or direction changers.

23.1 Simple Machines and their Purposes

A simple machine as a basic device that changes the magnitude or direction of a force to make work easier, emphasizing that it cannot create energy.

- **Load (L):** The resistance force or weight to overcome (in Newton).
- **Effort (E):** The force applied to the machine (in Newton).
- **Work Input (W_{in}):** The work done on the machine, equal to effort x distance moved by effort.
- **Work Output (W_{out}):** The useful work done by the machine on the load, equal to Load x distance moved by load.

The Three Primary Functional Purposes:

- **Force Multipliers (MA > 1):** Small effort moves a large load over a shorter distance (e.g., car jacks, nutcrackers).
- **Speed/Distance Multipliers (MA < 1):** Large force over a small distance moves a load a much greater distance/speed (e.g., broom, fishing rod, bicycle gears).
- **Direction Changers (MA = 1):** Alters only the direction of the force for convenience (e.g., flagpole pulley).

Mechanical Advantage MA: A number that indicates how many times the machine multiplies the input force.8 It is the ratio of the load to the effort.

Note: It has no units. If MA > 1, the machine makes it easier to lift heavy objects.

$$MA = \frac{\text{Load}}{\text{Effort}} = \frac{L}{E}$$

Velocity Ratio (VR): The ratio of the distance moved by the effort to the distance moved by the load. Unlike MA, VR is usually fixed by the geometry of the machine and is not affected by friction.

$$VR = \frac{\text{Distance moved by Effort}}{\text{Distance moved by Load}} = \frac{d_e}{d_l}$$

Efficiency (η): The measure of how much of the input work converted into useful output work. In the real world, efficiency is always less than 100% because some energy is lost to friction and heat.

$$\eta = \frac{Work\ Output}{Work\ Input} \times 100\% = \frac{MA}{VR} \times 100\%$$

Activity 23.1: From the following local tools (e.g., a car jack, broom, and flagpole pulley), identify whether each is a force multiplier, distance multiplier, or direction changer. Explain your answer by comparing the distance your hands move with the distance the load moves.

Example 23.1: A man uses a lever to lift a crate. He applies a force of 200 N to move a load of 600 N. If the distance moved by the effort is 1 m, and the distance moved by the load is 0.25 m, calculate: (a) V.R, (b) AMA and (c) Efficiency:

Solution:

a. $V = \frac{d_E}{d_L} = \frac{1\text{m}}{0.25\text{m}} = 4$

b. b. $MA = \frac{L}{E} = \frac{600\text{N}}{200\text{ N}} = 3$

c. c. $\eta = \frac{MA}{VR}$ x 100% = 75%

Exercise 23.1 A worker applies 250 N effort through a distance of 2 m to lift a 800 N load by 0.5 m. The efficiency of the machine is:

Review Questions for lesson 23

23.1. A student uses a machine to lift a heavy load. The load is 600 N, and the effort applied is 150 N. The mechanical advantage of the machine is:

A. 0.25 B. 2 C. 4 D. 750

23.2. Which one of the following simple machines mainly acts as a distance (speed) multiplier?

A. Car jack B. Nutcracker C. Bicycle D. Flagpole pulley

23.3. A machine requires an effort that moves 2.4 m to lift a load through 0.6 m. If the machine lifts a 480 N load using an effort of 160 N, the efficiency of the machine is:

A. 50% B. 60% C. 75% D. 80%

23.4. Which statement best explains why no real machine can have an efficiency greater than 100%?

A. Machines always increase the input energy.

B. Some input energy is lost as heat and friction.

C. Machines create additional energy while operating.

D. Mechanical advantage is always less than one.

Suggested Teaching Methods

- **Scaffolding**: Guide students from the baseline concepts of work (W = Fd) into the ratios for force multiplication (MA) and distance multiplication (VR).
- **Concept Mapping:** Map out the structural split between Ideal (theoretical) performance metrics and Actual (real-world) metrics.

Teaching Resources

- Core Material: Grade 9 text book and Grade 11 Tutorial Physics Condensed Curriculum.
- Demonstration Items: Rulers, markers, blocks, and string to model force vs. distance variations.

Expected Learning Outcomes

Students will be able to define, calculate, and explain the relationships between mechanical advantage, velocity ratio, and efficiency. They will also be able to evaluate how friction impacts real-world machine output.

Lesson 24(45 minutes)

Topic Simple Machine

Sub-topic: Classification and Mechanics of Simple Machines

At the end of this lesson delivery, students will be able to:

- Classify the three structural classes of levers based on the positions of the fulcrum, load, and effort.
- Calculate the MA, VR, and efficiency of inclined planes, wedges, and screws.
- Analyze wheel and axle systems to determine if they are configured as force multipliers or speed multipliers.
- Determine the velocity ratio of fixed, movable, and compound pulley systems.

24.1 The Lever

Levers: A rigid bar that rotates around a fixed pivot point called a fulcrum.

- **First Class**: Fulcrum sits between the effort and the load (e.g., seesaws, crowbars, etc.). Can multiply force or speed.
- **Second Class**: Load sits between the fulcrum and the effort (e.g., wheelbarrows, nutcrackers, bottle openers, etc.). Always acts as a force multiplier (MA > 1).
- **Third Class**: Effort sits between the fulcrum and the load (e.g., tweezers, fishing rods, the human forearm, etc.). Always acts as a speed/distance multiplier (MA < 1).

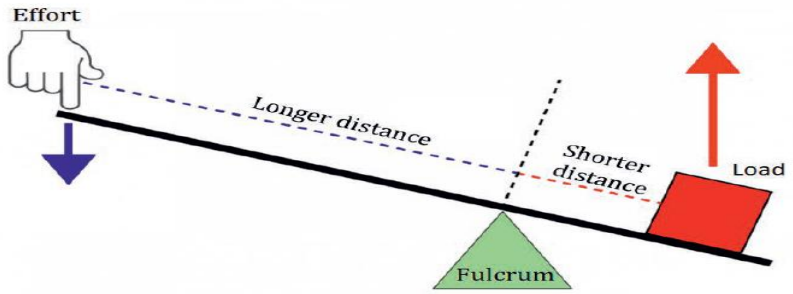


Fig 24.1 Feature of lever

Wheel and Axle: Consists of a large wheel attached to a smaller axle that rotate together.

- **Force Multiplier**: Effort is applied to the outer wheel to turn a heavy load on the axle (VR = > 1), e.g., screwdrivers, doorknobs).

$$\text{VR} = \frac{\text{distance moved by effort}}{\textit{distance moved by load}} = \frac{2\pi R}{2\pi r} = \frac{R}{r}$$

- **Speed Multiplier**: Effort is applied to the inner axle to spin the outer wheel faster (VR = <1, e.g., bicycle gears).

Pulleys: A grooved wheel running a cable system.

- **Fixed Pulley**: VR = 1. Only redirects force.
- **Single Movable Pulley**: VR = 2. Halves the required effort force.
- **Compound Pulley** (Block & Tackle): VR = number of rope segments supporting the moving load.

24.2 The Inclined Plane Family (Sloped Systems)

- **Inclined Plane**: A flat, sloped surface used to lift loads using less force over a longer distance.



Figure 14.2 Inclined plane

Mechanical Advantage (M.A) = MA = $\frac{\text{Load}}{\text{Effort}}$

Velocity ratio (VR) = $\frac{\text{length of the incline surface}}{\text{height of inclined plane}} = \frac{l}{h}$

Efficiency (η) = $\frac{MA}{VR}$ x 100%

Wedge: Two inclined planes placed back-to-back used to split materials apart.

Velocity ratio (VR) = $\frac{\text{penetration length}}{\text{thickness}} = \frac{l}{t}$

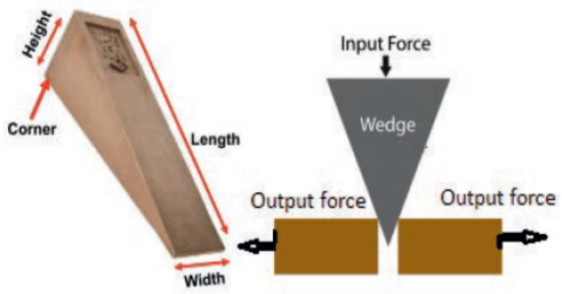


Fig 24.3 The wedge

Screw: An inclined plane wrapped around a central cylinder to form threads.

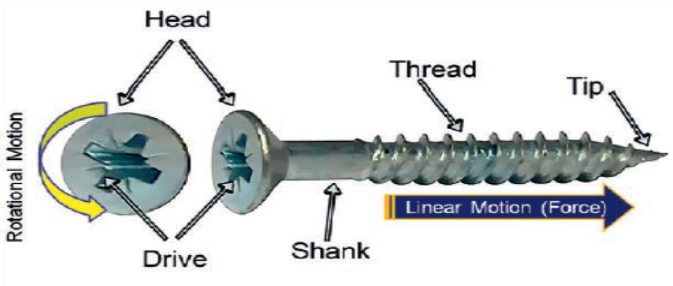


Fig 24.4

Velocity ratio = $\frac{\text{Circumference of rotation of the handle}}{pith} = \frac{\pi d}{P} = \frac{2\pi r}{P}$

Where **r** = is the radius of the turning handle and **p** = is the pitch (the distance between adjacent threads or p = $\frac{\text{length of the threads}}{number\ of\ threads}$).

Activity 24. 1: Act out lifting an object using a forearm, cutting with scissors, and wheeling a load. Identify where the pivot, input force, and resistance sit for each action to determine its lever class.

Example 24.1 A screw has 25 threads over a length of 50 mm. The handle radius is 8 cm. Determine the velocity ratio.

Solution:

p = $\frac{\text{length of the threads}}{number\ of\ threads} = \frac{50\text{mm}}{25}$ = 2mm = 0.002m

Velocity ratio = $\frac{2\pi r}{P} = \approx 251$

Exercise 24.1 single movable pulley is being used to move a 140 N load. The pulley is a little dirty, so it adds another 5 N of frictional force.

a) Can this load be moved with a 75 N input force? Explain your answer.

b) Would a single fixed pulley work? Explain your answer.

Review Questions for Lesson 24

24.1. Which one of the following simple machines always acts as a force multiplier?

A. Fishing rod
C. Wheelbarrow
B. Tweezers
D. Fixed pulley

24.2. Which statement correctly compares a fixed pulley and a single movable pulley?

A. Both are force multipliers.

B. Both have a velocity ratio of 2.

C. A fixed pulley changes only the direction of the effort, while a movable pulley reduces the effort needed.

D. A movable pulley has a velocity ratio of 1.

24.3. A wheel has a radius of 24 cm and is rigidly attached to an axle of radius 6 cm. The effort is applied to the wheel. The velocity ratio of the machine is:

A. 0.25 B. 2 C. 4 D. 30

24.4. An inclined plane is 5 m long and raises a load through a vertical height of 1 m. If the machine has a mechanical advantage of 4, its efficiency is:

A. 40% B. 60% C. 80% D. 125%

24.5. A wedge has a penetration length of 20 cm and a thickness of 4 cm. The velocity ratio of the wedge is:

A. 4 B. 5 C. 16 D. 24

24.6. A screw has 40 threads over a threaded length of 80 mm. The handle radius is 10 cm. The velocity ratio of the screw is closest to:

A. 31 B. 157 C. 314 D. 628

24.7. A student claims that a machine with a large mechanical advantage must always have a high efficiency. Which statement is correct?

A. The claim is always true.

B. The claim is false because efficiency also depends on the velocity ratio and friction.

C. Efficiency and mechanical advantage are unrelated.

D. Efficiency is always greater than the mechanical advantage.

Suggested Teaching Methods

Structural Visualizations: Use geometric diagrams to show how the wedge and screw are derived from the basic inclined plane.

Formula Derivation Scaffolding: Walk through the derivation of the screw equation to show that the input distance equals the circular path of the handle ($2\pi r$), while the output distance equals the pitch (p).

Teaching Resources

- **Textbook / Guide:** Grade 9 text book and Grade 11 Tutorial Materials.
- **Props:** A wood screw, a hand screwdriver, and a simple wedge block to show how rotational input forces convert into linear output forces.

Expected Learning Outcomes

Students will be able to categorize the six types of simple machines into their respective mechanical families, deduce velocity ratios from component dimensions, and solve multi-step problems for complex setups.

Weekly Assessment & Revision for lesson 21-24 (Every Friday)

Every Friday should be dedicated to **assessment and revision** of the concepts and skills covered during the week.

The primary purpose of the weekly assessment is to:

- Monitor students' attendance and participation.
- Evaluate students' understanding of the week's lessons.
- Identify learning gaps and misconceptions.
- Provide timely feedback and make instructional adjustments where necessary.

Prepare **25–30 assessment questions** using a variety of item types (e.g., true/false, multiple choice, short-answer, problem-solving or conceptual questions) as appropriate.

Suggested Time Allocation for Friday Assessment

- **30 minutes – Assessment:** Students complete the assessment independently under examination conditions.
- **15 minutes – Review and Feedback:** The teacher reviews the answers with the class, explains the correct solutions, addresses common misconceptions and provides constructive feedback.

Lesson 26(45minutes)

Topic: Heat Conduction and Calorimetry

Sub-topic: Temperature vs. Heat, Heat Transfer Mechanisms

By the end of this lesson, students will be able to:

- Differentiate between temperature (intensive property) and heat (extensive property).
- Define thermal equilibrium and state the First Laws of Thermodynamics.
- Calculate internal energy changes using the sign conventions of the First Law of Thermodynamics.
- Identify and explain the three heat transfer mechanisms: conduction, convection, and radiation.
- Solve linear, area, and volume thermal expansion problems;

26.1 Temperature vs. Heat

- **Temperature**: An intensive property measuring the average kinetic energy of constituent particles. Independent of mass; measured in Kelvin (K).
- **Heat (Q):** An extensive property representing total thermal energy in transit between systems due to a temperature gradient. Measured in Joules (J). Heat flows spontaneously from higher temperature zones to lower temperature zones.
- **Thermal Equilibrium**: A state where interacting bodies reach identical temperatures, resulting in zero net heat transfer.

The First Law of Thermodynamics states that the total energy of an isolated system is constant. For a closed system, the change in internal energy (ΔU) is equal to the difference between the heat (Q) supplied to the system and the work (W) done by the system on its surroundings.

- The law is represented by the following equation : $\mathbf{\Delta U = Q - W}$
- **Sign Conventions**: +Q = heat absorbed; -Q = heat released. +W = work performed by the system (expansion); -W = work performed on the system (compression).

Example 26.1. A gas is compressed by an external force performing 300 J of work *on* it, while it simultaneously sheds 100 J of heat to its surroundings. What is the change in its internal energy?

Solution: Work done *on* system: W = -300J. Heat released: Q = -100J.

$\Delta U = Q - W$, $\Delta U = -100 - (-300) = +200J$ (The internal energy increases by 200J).

26.2 Heat Transfer Mechanisms

Conduction: Thermal energy transfer through a material via direct atomic/molecular collisions without bulk movement of the matter itself. Dominant in dense solids (like metals).

Convection: Thermal energy transfer via the bulk movement of fluid molecules (liquids or gases) driven by density changes. Hotter fluids expand, become less dense, and rise, while cooler fluids sink.

Radiation: Energy transfer via electromagnetic waves (primarily infrared waves). This mechanism requires no physical medium and can travel through a vacuum.

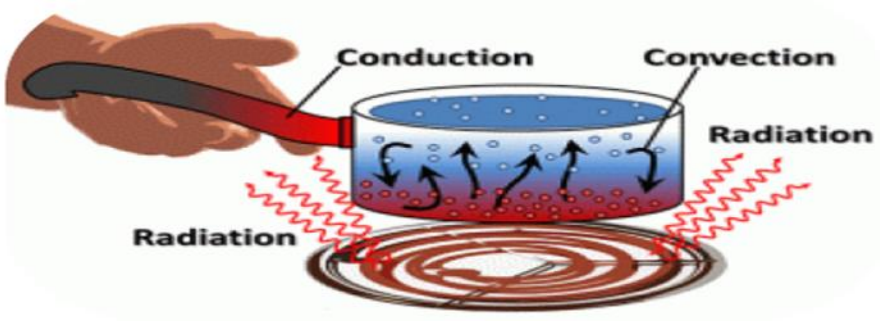


Fig 26.1 Summarizes the three heat transmission mechanism

Activity 26.1: For each of the following situations, determine the signs of heat (Q) and work (W), then predict whether the internal energy increases, decreases, or remains unchanged. If a gas is compressed by an external piston while losing heat to the surroundings.

26.3 Thermal Expansion (1D, 2D, and 3D)

Thermal expansion is the structural dimension increase of a substance when heated, driven by the expanding average separation between its constituent particles as their kinetic energy rises.

Linear Expansion (1D): Applies to long, thin elements like structural steel rails, rods, or wires.

The Formula**:** $\Delta L = \alpha L_0 \Delta T$

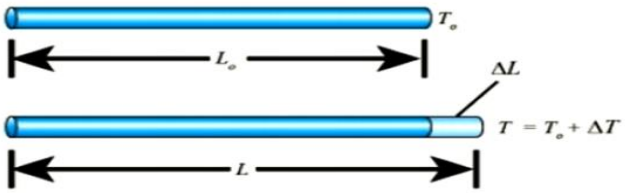


Fig 26.2 Linear expansion

Where α**(alpha**) is the coefficient of linear expansion (the unit $°C^{-1}$ or K^{-1}, L_0 is the initial length, and $\Delta T = T - T_0$ is the change in temperature

- **Real-World Impact**: Engineers insert expansion joints in concrete bridges to prevent buckling on hot summer days.

Area Expansion (2D): Governs flat plates or sheets. When heated, the length and the width both increase.

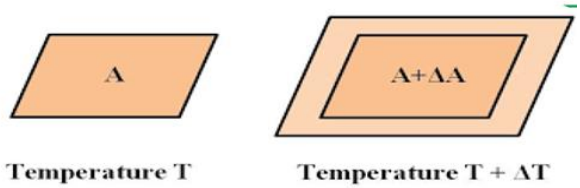


Figure 26.3 Area expansion

$$\Delta A = \beta A_0 \Delta T \quad (\text{where } \beta = 2\alpha \text{ for isotropic materials})$$

Note: If a metal sheet contains a hole, the hole expands too! It behaves exactly as if it were filled with the same metal.

Volume Expansion (3D): Governs three-dimensional solids, liquids, and gases.

$$\Delta V = \gamma V_0 \Delta T \quad (\text{where } \gamma = 3\alpha \text{ for isotropic solids})$$

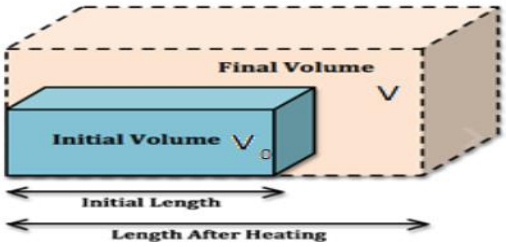


Fig 26.4 Volume expansion

Anomalous Behavior of Water (Between 0°C and 4°C).

- Heating from 0°C to 4°, water **contracts** when heated (volume drops, density increases.
- **Above 4°C:** It behaves normally (expands).
- Water has its minimum volume and maximum density at exactly **4°C**.

Liquids vs. Gases: Liquid volume expansion measurements yield **apparent values** because the heated container expands too.

Gases expand much more than condensed phases and are uniquely governed by both temperature and pressure.

Example 26.2. A steel bridge segment is exactly 50m long at 20ºC. If peak summer heat drives the temperature to 40ºC, by how many centimeters does it expand? α = 1.2 x 10^{-5} $^{o}C^{-1}$.

Solution: $\Delta L = \alpha L_0 \Delta T = 1.2cm$

Exercise 26.2. By how much does the volume of an aluminum cube 5.00 cm on an edge increase when the cube is heated from 10 ◦C to 60 ◦C? [Use $\alpha_{Al} = 23 \times 10^{-6}$/ C◦.]

26.4 Phase Changes & Diagram Structure

- **Phase Transitions**: A transformation between solid, liquid, or gas states that proceeds at a constant temperature for a given pressure.
- A **phase diagram** is a pressure-temperature (P-T) graph mapping the thermodynamic stability boundaries of a substance:
- **Phase Lines:** Representation curves showing two phases coexisting in stable equilibrium (Melting/Freezing line, Vaporization/Condensation line, Sublimation/Deposition line).
- **Triple Point**: The unique intersection where solid, liquid, and gas phases stably coexist simultaneously.
- **Critical Point:** The terminal point of the liquid-gas boundary, beyond which the substance forms an indistinguishable supercritical fluid.

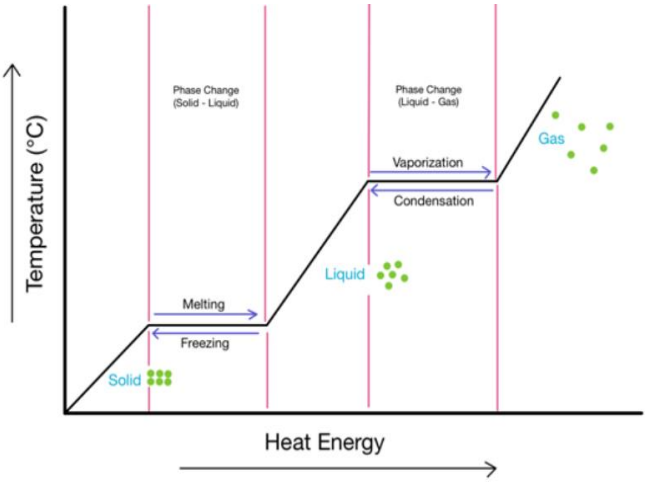


Fig 26.5Temperature versus heat diagram

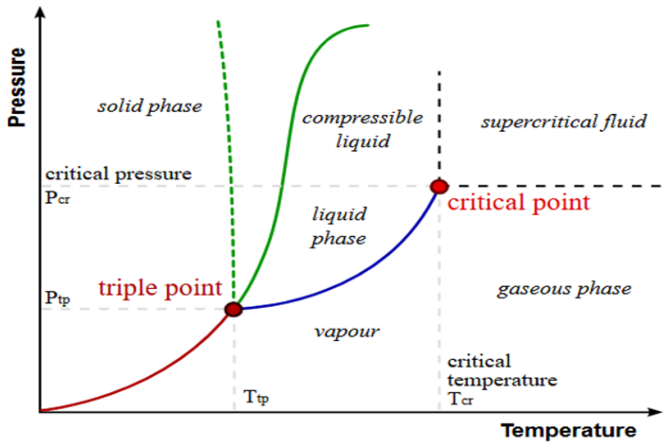


Fig 26.6 Phase diagram

Review Questions for Lesson 26

26.1. Two identical metal blocks, A and B, are made of the same material. Block A has a mass of 1 kg at 80°C, while block B has a mass of 2 kg at 80°C. Which statement is correct?

A. Both blocks contain the same amount of heat because their temperatures are equal.

B. Block B has greater thermal energy than block A because heat is an extensive quantity.

C. Block A has more heat because it has less mass.

D. Temperature depends on the mass of the object.

26.2. A gas absorbs 600 J of heat while expanding and performing 250 J of work on the surroundings. What is the change in the internal energy of the gas?

A. +850 J B. +350 J C. -350 J D. -850 J

26.3. A gas is compressed by an external piston while releasing heat to the surroundings. The magnitude of the work done on the gas is 500 J, and the heat released is 200 J. According to the First Law of Thermodynamics, the change in internal energy is:

A. +300 J B. +700 J C. -300 J D. -700 J

26.4. Two identical containers hold equal masses of water and cooking oil, initially at the same temperature. Equal amounts of heat are supplied to each. Which statement is most likely to be **correct?**

A. Water reaches a higher temperature because it is denser.

B. Oil reaches a higher temperature because its specific heat capacity is lower than that of water.

C. Both reach the same temperature because they receive equal heat.

D. Water reaches a higher temperature because it has a higher specific heat capacity.

26.5. The following diagrams show the three methods of heat transfer between sources of heat and objects.

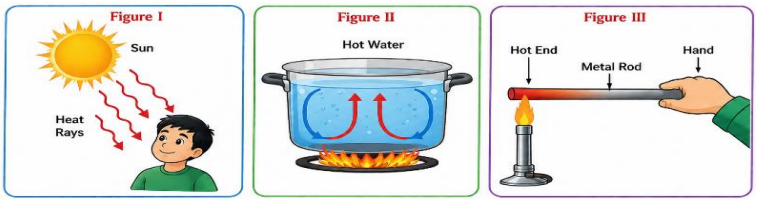


Fig 26.2

Which of the following is the **correct** method of heat of transfer between sources of heat and objects of figure I, II and III respectively?

A. Radiation, convention and conduction

B. conduction, radiation and convention

C. Convention, conduction and radiation

D. radiation, conduction and convention

26.6. An engineer designs a steel bridge without leaving expansion joints between adjacent sections. What is the most likely consequence during extremely hot weather?

A. The bridge contracts and develops gaps.

B. The bridge expands, producing compressive forces that may cause buckling.

C. The bridge melts because steel reaches its melting point.

D. The bridge experiences no dimensional change because steel is a solid

26.7. A brass rod and an iron rod have the same initial length and are heated through the same temperature rise. The brass rod expands more than the iron rod. This is because brass has

A. a larger density.

B. a higher melting point.

C. a lower specific heat capacity.

D. a larger coefficient of linear expansions

26.8. A metal of length 18.02m is heated until its temperature rises to 80 0C. If its new length is 20.01m, calculate its linear expansively.

26.9. You are building a plane, but the stainless steel rivets you are using will not fit through the holes in the skin of the plane. At 20°C, each rivet has a diameter of 12.020 mm, while the diameter of a hole is 12.000 mm. To what temperature should you cool a rivet so it fits in a hole? Use $\alpha_{\text{stainless steel}} = 1.7 \times 10^{-5}/\,C^{\circ}$.]

Suggested Teaching Methods

- **Real-World Contextualization:** Use everyday situations (like boiling water or the rapid heating of metal spoons) to anchor heat transfer concepts.
- **Sign Convention Practice:** Use quick card drills or hand signals to help students master the plus/minus signs for heat and work before they solve equations.

Teaching Resources

- **Core Material:** Grade 9- 11 Physics Tutorial Syllabus.
- **Visual Aids:** Diagrams or charts illustrating conduction, convection currents, and radiation waves.

Expected Learning Outcomes

Students will be able to distinguish between thermal state parameters, apply the laws of thermodynamics to compute energy shifts, identify active heat transfer modes, and compute thermal mass values using calorimetry equations.

Lesson 27(45minutes)

Topic: Heat Conduction and Calorimetry

Sub-topics: Specific Heat Capacity (Solid & Liquid)

After completing this section, you should be able to:

- Solve numerical problems using Q = mcΔT and energy conservation principles.
- Describe the experimental setups (electrical and mixture methods) for determining the specific heat capacity of solids and liquids.
- Explain real-world applications of specific heat capacity in everyday life.

27.1 Measuring specific heat capacity

- **Calorimetry** measures heat exchanged based on the Law of Conservation of Energy: Heat Lost = Heat Gained.
- **Specific heat capacity (c**) is the heat required to raise 1kg of a substance by 1°C (or 1 K).

$$Q = mc\Delta T$$

➢ There are two common methods of measuring specific heat capacity: Electrical method and Method of mixtures

Electrical methods:

- For electrical heating methods, the energy supplied by the heater is accurately determined from the measured voltage(**V**), current(**I**), and heating time (**t**): **E= V It**.
- To determine the **specific heat capacity of a solid**, assume no heat loss so that the electrical energy supplied by the heater **equals t**he heat gained by the metal block.

VI t = mcΔT

Specific heat capacity of a solid (**c**) $= \frac{\mathbf{VIt}}{\mathbf{m\Delta T}}$

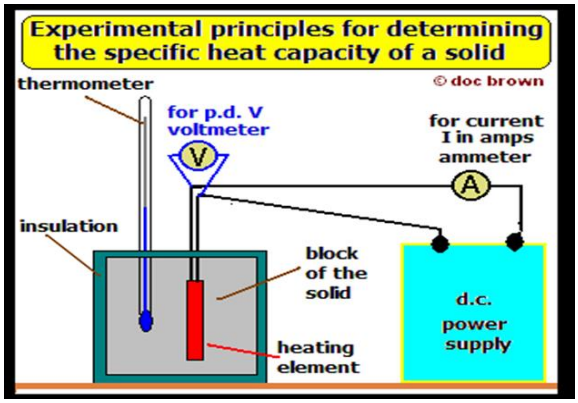


Fig 27.1 Finding the specific heat capacity of a solid by electrical method

Example 27.1. 98,000J of electrical heat are needed to raise the temperature of 2kg of a substance from 51°C to 65°C. Calculate the specific heat capacity of a substance.

Solution: VI t = mcΔT; $c = \frac{VIt}{m\Delta T} = \frac{98000J}{2kg\,(65-51)oC}$ = 3500 J/kg ^{o}C

Exercise 27.1. A heater rated 2KW is used for heating the solid of mass 6kg, if its temperature rises from 30℃ to 40℃. In 12 seconds, find the S.H.C of the solid.

27.2 Determining the Specific Heat Capacity of a Liquid (Electrical Method)

- Place a liquid of mass m_L in a calorimeter (mass m_c, specific heat capacity c_c) and heat it electrically for time **t** using a heater of voltage **V** and current **I.**
- Measure the temperature rise from T_i to T_f. Assuming no heat loss,

Electrical energy supplied = Heat gained by liquid + heat gained by calorimeter

$$\mathbf{V\,It = m_L\,c_L\,(T_f - T_i) + m_c\,c_c\,(T_f - T_i)}$$

Hence, the specific heat capacity of the liquid is:

$$c_L = \frac{VIt - m_c\,c_c\,(T_f - T_i)}{m_L\,(T_f - T_i)}$$

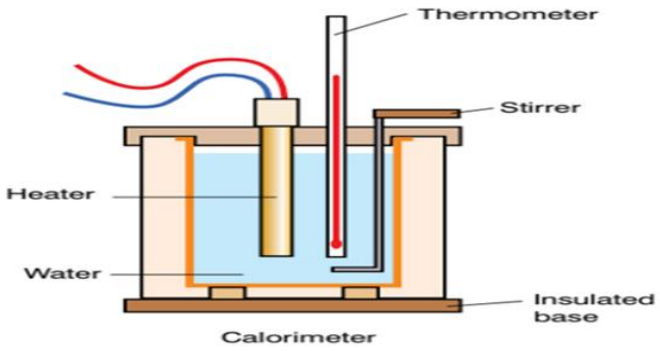


Fig 27.2 Determining the specific heat capacity of a liquid by electrical method.

Example 27.2. A 0.25 kg liquid is heated electrically in a 0.10 kg copper calorimeter. A 12 V, 5 A heater operates for 10 minutes. The temperature rises from 20°C to 45°C. Find the specific heat capacity of the liquid. Use c_{Cu} = 390 J/kg °C; c_w = 4200 J/kg °C.

Solution:

$c_L = \frac{VIt - m_c\,c_c\,(T_f - T_i)}{m_L\,(T_f - T_i)}$ = 5604 J/kg °C.

Exercise 27.2. A 0.30 kg sample of water is heated electrically in a 0.15 kg copper calorimeter. The initial temperature of both the water and the calorimeter is 20°C, and after heating, the final temperature is 50°C. Find the total heat absorbed. Use: c_{Cu} = 390 J/kg °C; c_w = 4200J/kg °C:

27.3 Method of Mixtures (Determining the Specific Heat Capacity of a Solid)

- Heat a solid of mass $\mathbf{m_S}$ to temperature $\mathbf{T_h}$.
- Transfer it quickly into a calorimeter (mass $\mathbf{m_c}$, specific heat $\mathbf{c_c}$) containing water (mass $\mathbf{m_w}$) at temperature $\mathbf{T_c}$.
- Stir until the final equilibrium temperature $\mathbf{T_f}$ is reached.
- In an ideal, insulated system, the energy exchange follows the Law of Conservation of Energy:

Heat lost by hot solid body = Heat gained by water + Heat gained by calorimeter

$$m_S\, c_S\, (T_h - T_f) \quad = \quad m_w\, c_w\, (T_f - T_c) + m_c\, c_c\, (T_f - T_c)$$

Hence

$$\mathbf{c_S = \frac{m_w\, c_w\, (T_f - T_c) + m_c\, c_c\, (T_f - T_c)}{m_S\, (T_h - T_f)}}$$

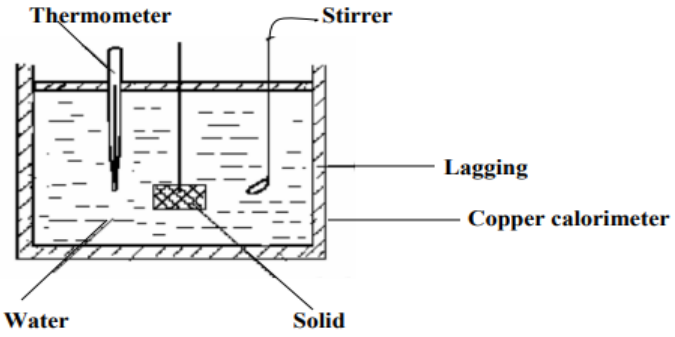


Fig 27.3 Determining the specific heat capacity of a solid using the method of mixtures

Example 27.3 A 0.20 kg metal block at 100°C is placed into 0.40 kg of water at 20°C contained in a 0.10 kg copper calorimeter. The final temperature is 25°C. Find the specific heat capacity of the metal. Use: $\boldsymbol{c_{Cu}}$ = 390 J/kg °C; $\boldsymbol{c_w}$ = 4200 J/kg °C:

Solution: T_h= 100°C; T_f =25°C; T_c =20°C;

$$c_S = \frac{m_w\, c_w\, (T_f - T_c) + m_c\, c_c\, (T_f - T_c)}{m_S\, (T_h - T_f)} = 573 \text{ J/kg °C}$$

Exercise 27.3. The temperature of a piece of copper of mass 250g is raised to 100℃ and it is then transferred to a well- lagged aluminum can of mass 10.0g containing 120g of methylated spirit at 10.0℃. Calculate the final steady temperature after the spirit has been well stirred. Neglect the heat capacity of the stirrer and any losses from evaporation. (S.H.C of copper, aluminum and spirit respectively $\mathbf{c_{Cu}}$ = **400** $\boldsymbol{J}$**/kg** $\boldsymbol{k}$ **,** $\mathbf{c_{Al}}$ **=900** $\boldsymbol{J}$**/kg** $\boldsymbol{k}$ $\mathbf{c_{Sprite}}$ **= 2400** $\boldsymbol{J}$**/kg** k).

Review Questions for lesson 27

27.1. Two metal blocks, **A** and **B**, have equal masses. When supplied with the same amount of heat, the temperature of block A increases by 20°C, while that of block B increases by 10°C. Which statement is correct?

A. The specific heat capacity of A is twice that of B.

B. The specific heat capacities of A and B are equal.

C. The specific heat capacity of B is twice that of A.

D. The heat absorbed by B is twice that absorbed by A.

27.2. A 200 W heater warms a 2.0 kg metal block for 5 minutes. The temperature rises from 25°C to 40°C. Assuming no heat loss, the specific heat capacity of the metal is:

A. 1000 J/kg °C B. 1500 J/kg °C C. 2000 J/kg °C D. 3000 J/kg °C

27.3. In an electrical calorimetry experiment, why must the heat absorbed by the calorimeter be included when calculating the specific heat capacity of a liquid?

A. The calorimeter loses heat to the surroundings.

B. The calorimeter absorbs part of the electrical energy supplied.

C. The calorimeter increases the heater's power.

D. The calorimeter changes the mass of the liquid.

27.4. Which of the following practices minimizes experimental error when determining the specific heat capacity of a hot metal by the method of mixtures?

A. Allow the metal block to cool before transferring it.

B. Use a large open container instead of a calorimeter.

C. Measure the initial temperature after mixing.

D. Stir the mixture continuously and transfer the metal quickly.

27.5. A student determines the specific heat capacity of a metal using the electrical method but forgets to include the heat absorbed by the calorimeter. The calculated value of the specific heat capacity of the metal will be:

A. greater than the true value. B. less than the true value. C. unchanged. D. impossible to determine.

Suggested Teaching Methods: Cooperative learning, step-by-step problem-solving modeling, and structured diagram analysis.

Teaching Resources: Grade 11Textbook, Figures 27.2 & 27.3, immersion heater, calorimeter cup, thermometer, stirrer, power supply.

Expected Learning Outcomes: Students will be able to set up heat balance equations for mixed or electrically heated systems and solve for unknown specific heat capacities of a substance.

Lesson 28 (45 minutes)

Topic: Heat Conduction and Calorimetry

Sub-topic: Measuring Specific Latent Heat (Hidden heat)

After completing this section, you should be able to:

- Define specific latent heat.
- Distinguish between latent heat of fusion and vaporization.
- Apply the principle of conservation of energy to determine latent heat.
- Explain real-world applications of latent heat in everyday life.

28.1 Specific Latent Heat(L)

- **Specific latent heat (L):** is the energy required to change the state of 1 kg of a substance **without changing** its temperature. There are two types of latent heat namely:
- **Latent heat of fusion (L_f)**: Solid → Liquid at constant temperature.
- **Latent heat of vaporization (L_v)**: Liquid → Gas at constant temperature.

Heat required = mass x specific latent heat of fusion

$$\mathbf{Q = mL}$$

Where Q = heat (J), m = mass (kg), and L = specific latent heat (J /kg)

Note: (L_V) is much larger than **(L_f)** because completely separating molecules into a gas requires breaking all intermolecular bonds.

28.2 Measuring the Latent Heat of Fusion of Ice (Mixture Method)

- A method used to determine the latent heat of fusion of ice by applying the principle of conservation of energy.
- When ice is mixed with warm water,

Heat lost by warm water + calorimeter = Heat gained by the ice

➢ The ice gains heat to:

- Melt (needs latent heat of fusion)
- Warm from 0°C to the final temperature.

➢ Energy Balance Equation:

Heat lost by warm water + calorimeter = Heat used to melt ice + heat to warm melted ice

$$\mathbf{m_W\ c_W\ (\ T_1\ -\ T_{f)}\ +\ m_c\ c_c\ (\ T_1\ -\ T_{f)}\ = m_{ice}\ L_f + m_{ice}\ c_W\ (\ T_f\ -\ 0°C)}$$

Where:

m_W = mass of warm water, m_c = mass of calorimeter, m_{ice} = mass of ice

c_W= specific heat capacity of water, c_c = specific heat capacity of calorimeter

T_1= initial temperature of warm water

T_f = final equilibrium temperature of mixture

L_f = latent heat of fusion of ice

Hence, the latent heat of fusion of ice is:

$$\mathbf{L_f = \frac{m_w\ c_w\ (T_1 - T_f) + m_c\ c_c\ (T_1 - T_f) - m_{ice}\ c_w\ T_f}{m_{ice}}}$$

Example 28.1. A 0.30 kg sample of water at 40°C is contained in a 0.10 kg copper calorimeter $\mathbf{c_c}$=390 J/kg °C. A 0.08 kg piece of ice at **0°C** is added. The final temperature is 15°C. Find the specific latent heat of fusion of ice. Take: c_w = 4200 J/kg °C.

Solution: $L_f = \frac{m_w\ c_w\ (T_1 - T_f) + m_c\ c_c\ (T_1 - T_f) - m_{ice}\ c_w\ T_f}{m_{ice}} = 3.45\ x10^5$ J/kg

Exercise 28.1: A 0.20 kg sample of water at 40°C is contained in a 0.10 kg copper calorimeter $\mathbf{c_c}$=390 J/kg °C. A 0.05 kg piece of ice at 0°C is added. Assume the latent heat of fusion of ice is $L_f = 3.45\ x10^5$ J/kg. Find the final equilibrium temperature of the mixture. Take: c_w = 4200 J/kg °C.

28.3 Measuring the specific latent heat of steam

- This method is used to determine the latent heat of vaporization of water by measuring the heat required to convert water into steam at its boiling point.
- When steam at 100 ∘C enters cooler water in a calorimeter:

Heat lost by steam = Heat gained by water + calorimeter

- The steam releases heat in two stages**:**
  - Condensing of steam to water at 100 ◦C (latent heat released): $Q_1 = m_S\ L_V$
  - Cooling of condensed steam from 100 ∘C to the final temperature: $Q_2 = m_S\ c_W\ (100\text{-}T_f)$
- Energy balance equation:

$$\mathbf{m_S\ L_V\ +\ m_S\ c_W\ (100\ -T_f\ ) = m_W\ c_W\ (T_f - T_1) + m_C\ c_c\ (T_f - T_1)}$$

Where:

m_S= mass of steam; m_W = mass of water; m_C = mass of calorimeter

L_V = latent heat of vaporization

c_W= specific heat capacity of water

c_c= specific heat capacity of calorimeter

T_1 = initial cooler temperature of water and calorimeter

T_f = final equilibrium temperature of mixture

Hence, the latent heat of vaporization of steam is:

$$\mathbf{L_V = \frac{m_w\, c_w\, (T_f - T_1) + m_c\, c_c\, (T_f - T1) - m_S\, c_w\, (100\,^{\circ}C - T_f)}{m_S}}$$

- Daily Life Applications of Latent Heat:
  - Cooking with steam
  - Cooling drinks with ice
  - Steam power plants
  - Cloud formation and rainfall
  - Fire extinguishing
  - Food preservation

Example 28.2. A 0.30 kg sample of water at 20°C is contained in a 0.10 kg copper calorimeter $\mathbf{c_c}$=390 J/kg °C. Steam at 100°C is passed into the calorimeter until the final temperature becomes 50°C. The mass of condensed steam is 0.025 kg. Find the specific latent heat of vaporization of steam. Take: c_w = 4200 J/kg °C.

Solution:

Heat gained by water: $Q_w = m_w\, c_w\, (T_f - T_1)$ =37800 J

Heat gained by calorimeter: $Q_c = m_c\, c_c\, (T_f - T_1)$ = 1170 J

Total heat gained: $Q_{gained} = Q_w + Q_c$ = 38970 J

- Heat released by steam:
  - Condensation: $Q_1 = m_S\, L_V = 0.025\, L_V$
  - Cooling from 100°C to 50°C: $Q_2 = m_S\, c_W\, (100 - T_f)$ = 5250 J
- Energy balance:

$0.025\, L_V + 5250 = 38970$ J

$L_V = 1.35 \times 10^6$ J/kg

Exercise 28.2: A 0.25 kg sample of water at 25°C is contained in a 0.10 kg aluminum calorimeter c_{Al} = 900 J/kg °C. 0.020 kg of steam at 100°C is passed into the calorimeter. Assume the specific latent heat of vaporization of steam is $L_V = 2.26 \times 10^6$ J/kg. Find the final equilibrium temperature. Take: c_w = 4200 J/kg °C.

Review Questions for Lesson 28

28.1. A student continuously heats a block of ice at **0°C** until it completely melts. During melting, the temperature of the ice-water mixture remains constant even though heat is supplied. The best explanation is that the supplied heat is used to:

A. increase the kinetic energy of the water molecules.

B. increase the potential energy by overcoming intermolecular forces.

C. raise both the kinetic and potential energies equally.

D. increase the pressure inside the mixture.

28.2. In an experiment to determine the latent heat of fusion of ice, the ice added is **wet** instead of perfectly dry. The calculated value of the latent heat of fusion will most likely be:

A. greater than the true value.

B. unchanged.

C. less than the true value.

D. impossible to predict.

28.3. During an experiment to determine the latent heat of vaporization of steam, some steam escapes without condensing in the calorimeter. The calculated value of L_v will be:

A. greater than the true value.

B. unchanged.

C. less than the true value.

D. equal to zero.

28.4. Which everyday application depends primarily on the large latent heat of vaporization of water?

A. A metal spoon becoming hot in tea.

B. Human body cooling through sweating.

C. Ice keeping drinks cold.

D. Expansion of railway tracks on hot days.

28.5. A 0.3 kg sample of water at 20°C is contained in a 0.1 kg copper calorimeter c_{Cu} =390 J/kg °C. Steam at 100°C is passed into the calorimeter until the final temperature becomes 50°C. The mass of condensed steam is 0.025 kg. The specific latent heat of vaporization of steam is closest to Take: c_W = 4200 J/kg °C.

A. 1.35×10^6 J kg

B. 1.8×10^6 J /kg

C. 2.26×10^6 J/ kg

D. 2.8×10^6 J/ kg

28.6. A 0.25 kg sample of water at 25°C is contained in a 0.10 kg aluminum calorimeterc_{Al} = 900 J/kg °C. 0.020 kg of steam at 100°C is passed into the calorimeter. Assume the specific latent heat of vaporization of steam is 2.26×10^{6} J/ kg. The final equilibrium temperature is approximately. Take: c_W = 4200 J/kg °C.

Suggested Teaching Methods: Conceptual mapping of heat pathways, contextual learning via real-life applications, guided mathematical derivation.

Teaching Resources: Grade 11Textbook, safety instructions for steam handling, boiling apparatus, ice blocks.

Expected Learning Outcomes: Students will accurately distinguish between sensible heat ($mc\Delta T$) and latent heat (mL) in complex thermal systems and solve related multi-step calorimetry equations.

Lesson 29(45 minutes)

Topic: Mechanical Oscillations and Sound waves

Sub-topic: Wave Propagation and characteristics, Mechanical Oscillations

After completing this section, you should be able to:

- Describe the fundamental principles of mechanical oscillations, restoring forces, and energy exchange.
- Determine the period of a simple pendulum based on string length and gravity.
- Analyze the dynamics, energy transfers, and period of a horizontal, frictionless spring-mass system using Hooke's Law.
- Explain the common characteristics of waves such as period, frequency, wavelength and amplitude

29.1 Wave Propagation and characteristics

Waves transfer energy without permanent matter displacement. Common characteristic of waves is described by:

- **Displacement(x)**: is the distance covered by a vibrating body at any time from its equilibrium (rest) position.
- **Amplitude (A)**: Maximum displacement of particles from equilibrium
- **Wavelength (λ):** Distance between two successive crests or compressions
- **Period (T):** The time taken for one complete wave or one cycle of the wave to pass a given point. Unit: seconds (s).
- **Frequency (f):** Number of waves passing a point per second (Hz). Frequency is the reciprocal of time period i.e. $f = \frac{1}{T}$.
- **Wave Speed (v):** Speed at which the wave travels
- **Wave Equation**: $v = f\lambda$

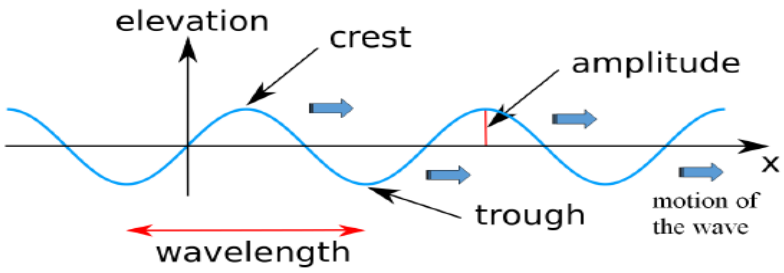


Figure 29.1 characteristics of wave

Mechanical Waves*:* are waves that require a material medium to travel. But cannot travel in a vacuum. Examples: Waves on a string, Water waves, Sound waves, etc**.**

Types of Mechanical Waves:

- **Transverse waves***:* Particles vibrate perpendicular to wave propagation (Crests/Troughs). E.g., strings, water.
- **Longitudinal waves:** Particles vibrate parallel to propagation (Compressions/Rarefactions). E.g., sound wave.

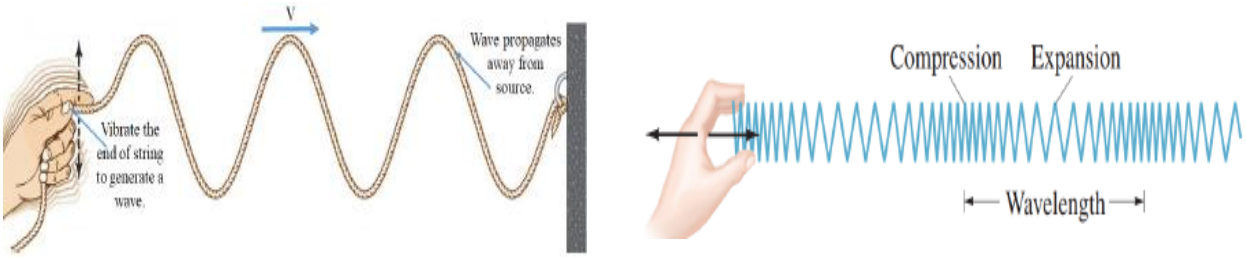


Fig 29.2 a wave on a string b. Compressions and rarefactions in a longitudinal wave

Exercise 29.1: A wave has a frequency of 500 Hz and a wavelength of 0.4 m. Find the speed of the wave.

29.2 Mechanical Oscillations and periodic motion

Mechanical oscillation: is a specific type of periodic motion where a body moves to-and –fro (back-and-forth) repeatedly about a fixed central position called the mean or equilibrium position

- Driven by a restoring force (like **gravity or elasticity**) that always pulls the object back toward the equilibrium position.
- Examples: a swinging pendulum, a spring–mass system, a vibrating ruler, and waves on water, etc.

Periodic motion: Any motion that repeats itself at regular intervals of time (called the period, T).

- It **does not require** an equilibrium point.
- Examples: earth orbiting the sun, hands of a clock moving in a circle, a spinning fan blade.
- **Note:** all oscillatory motions are periodic, but not all periodic motions are oscillatory.

Simple pendulum: A point mass **m** (the bob) suspended or attached from a fixed supported by a light string of length **L**.

- When displaced and released, it swings back and forth to wards equilibrium point by **restoring force of gravity**.

- This continuous oscillating the pendulum, continuous exchange of potential and kinetic energy.

➢ As long as the displacement angle remains small, the time period(T) of a simple pendulum is given by:

$$T = 2\pi \sqrt{\frac{L}{g}}$$

Where T = period(s); L = length of the string (m); g = acceleration due to gravity (m/s^2).

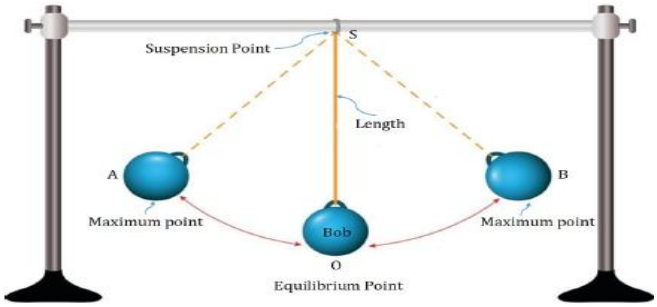


Fig 29.3 Simple pendulum

Key Insights:

- **Independent of mass m**: a heavy bob and a light bob take the same time to complete a swing.
- **Independent of amplitude**: valid as long as the displacement angle remains small ($\leq 15^o$).
- **T** $\propto \sqrt{\boldsymbol{L}}$; slower swing where string is longer.
- **T** $\propto \frac{\mathbf{1}}{\sqrt{\boldsymbol{g}}}$; swing faster where gravity is stronger.

Example 29.1. What is the period of a simple pendulum with length 30 cm? Use the acceleration due to gravity g = 10m/s^2.

Solution: $T = 2\pi \sqrt{\frac{L}{g}}$= 1.088sec

Exercise 29.2. What is the length of a simple pendulum with a time period of 2 seconds?

A spring- mass system (Horizontal, Frictionless)

- A block of mass **m** attached to an ideal spring with spring constant **k (stiffness**) in a frictionless horizontal surface that exchange energy between elastic potential and kinetic energy.
- **Restoring force ($\boldsymbol{F_{res}}$):** governed by Hooke's law that pulls the spring back toward its equilibrium position.

Hook's law states that: the force applied to elongate or compress a spring is directly proportional to its displacement from the equilibrium position, provided the elastic limit is not exceeded.

$$F_{res} = -kx$$

Where F_{res}: restoring force(N); k: spring constant or stiffness(N/m) and x: displacement from equilibrium point(m).

Note: the negative sign shows that the restoring force always acts in the opposite direction of the displacement of the object.

- The time period of a mass-spring oscillating system is:

$$T = 2\pi\sqrt{\frac{m}{k}}\ ;\ m = \frac{T^2\ k}{4\pi^2} \quad \text{and} \quad k = \frac{4\,\pi^2\ m}{T^2}$$

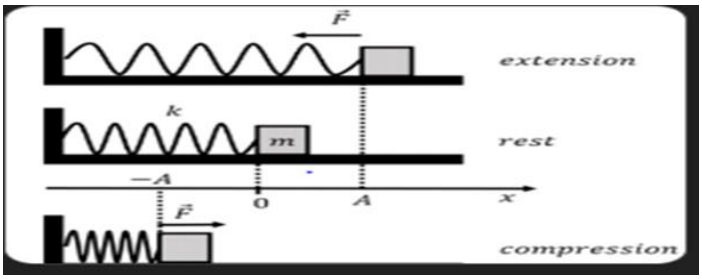


Fig 29.4 A mass - spring system undergoes periodic motion about x = 0.

Note: T: depends only on mass (m) and stiffness (k). Independent of amplitude, shape, or gravity.

Activity 29.1: Using the motion of a mass–spring oscillation system horizontally with friction less, identify the values of displacement, velocity, acceleration, kinetic energy, and elastic potential energy at maximum extension (+A) to maximum compression (−A).

Example 29.2: A 0.80 kg block oscillates with a period of 0.40 s. Find the spring constant.

Solution: $k = \frac{4\,\pi^2\ m}{T^2} = 197$ N/m

Exercise 29.3: An unknown block is attached to the same spring in two separate experiments.

Experiment 1: Mass = 1 kg; Period = 0.40 s

Experiment 2: Unknown mass =? Period = 0.80 s. Find the unknown mass.

Review Questions for lesson 29

29.1. Which one of the following statements correctly distinguishes oscillatory motion from periodic motion?

A. Every oscillatory motion is circular motion.
B. Oscillatory motion always requires a changing speed, whereas periodic motion always has constant speed.
C. Every periodic motion has a restoring force acting toward an equilibrium position.
D. Every oscillatory motion is periodic, but some periodic motions do not oscillate about an equilibrium position.

29.2. Two identical simple pendulums are taken to different planets. Planet X has g =16 m/s^2; Planet Y has g = 4 m/s^2. Ignoring air resistance, which statement is correct?

A. The pendulum oscillates twice as fast on Planet Y.
B. The period on Planet Y is twice that on Planet X.
C. The period is the same on both planets because the pendulums have equal masses.
D. The period depends only on the amplitude.

29.3. A block attached to a horizontal spring oscillates without friction. At the instant when the block passes through the equilibrium position,

A. Speed is maximum, restoring force is zero and kinetic energy is maximum.
B. Speed, acceleration and elastic potential energy are all maximum.
C. Potential energy and restoring force are maximum.
D. Speed is zero but kinetic energy is maximum.

29.4. A spring–mass oscillator has period T. The attached mass is doubled while the spring constant is increased fourfold. The new period becomes:

A. 2T B. ½ T C. $\frac{T}{\sqrt{2}}$ D. $\sqrt{2}$T

29.5. A wave has: Frequency = 20 Hz and Wavelength = 0.50 m. The wave enters another medium where its speed decreases to 6 m/s. Which statement is correct?

A. Frequency becomes 12 Hz while wavelength remains 0.50 m.
B. Frequency remains 20 Hz while wavelength becomes 0.30 m.
C. Both frequency and wavelength decrease.
D. Frequency decreases while wavelength increases.

29.6. Four systems are observed:

I. A child swinging on a swing through a small angle.

II. A satellite orbiting Earth.

III. A block oscillating on a frictionless spring.

IV. A vibrating tuning fork.

Which systems can be approximated as simple oscillatory motion?

A. I and II only
B. II and III only
C. I, III and IV only
D. I, II, III and IV

29.7. A block attached to a spring moves from maximum extension toward equilibrium. During this motion,

A. Kinetic energy decreases while elastic potential energy increases.
B. Both kinetic and elastic potential energies decrease.
C. Elastic potential energy is converted into kinetic energy.
D. Mechanical energy continuously decreases.

29.8. Which change does **not** affect the period of an ideal horizontal spring–mass oscillator?

A. Increasing the attached mass
B. Using a stiffer spring
C. Reducing the spring constant
D. Doubling the amplitude

29.9. A wave source produces waves at constant frequency. The wave enters another medium where its speed doubles. Which statement is **correct?**

A. Frequency doubles.
B. Wavelength doubles.
C. Period doubles.
D. Both wavelength and frequency double.

29.10. Whenever an object moves fastest, the restoring force acting on it must also be greatest. Which response is scientifically **correct**?

A. The statement is always true.
B. It is true only for pendulums.
C. It is true only for spring oscillators
D. It is false because restoring force is greatest where the speed is zero.

Suggested Teaching Methods: Comparative tabular analysis, direct derivation modeling, and interactive conceptual questions.

Teaching Resources: grade 9-10Textbook, Figures 29.1 & 29.2, pendulum bob with string, horizontal spring-mass demonstration apparatus.

Expected Learning Outcomes: Students will be able to set up restoring force equations, calculate periods for pendulums and spring systems, and identify positions of maximum kinetic and potential energy.

Weekly Assessment & Revision for lesson 26-29 (Every Friday)

Every Friday should be dedicated to **assessment and revision** of the concepts and skills covered during the week.

The primary purpose of the weekly assessment is to:

- Monitor students' attendance and participation.
- Evaluate students' understanding of the week's lessons.
- Identify learning gaps and misconceptions.
- Provide timely feedback and make instructional adjustments where necessary.

Prepare **25–30 assessment questions** using a variety of item types (e.g., true/false, multiple choice, short-answer, problem-solving or conceptual questions) as appropriate.

Suggested Time Allocation for Friday Assessment

- **30 minutes – Assessment:** Students complete the assessment independently under examination conditions.
- **15 minutes – Review and Feedback:** The teacher reviews the answers with the class, explains the correct solutions, addresses common misconceptions and provides constructive feedback.